A General Approach for Assessing Sensitivity to Violations of Causal Assumptions

Guilherme Duarte

Harvard University, Cambridge, Massachusetts 02138, USA

(gjduarte@fas.harvard.edu).

Abstract

Although causal sensitivity analyses are widely used to assess the robustness of conclusions, a key limitation is that existing methods are typically developed for specific designs. This paper develops a general sensitivity analysis framework that applies to a wide range of discrete causal models and estimands by deriving sharp bounds on causal estimands as a function of the maximum proportion of the population for which the assumption is relaxed. The framework is built on the generalized principal strata approach, which allows the translation of assumptions, estimands, and data into polynomial programs. In this approach, three classes of assumptions can be relaxed: a) functional restrictions, which limit the ways that one variable can respond to another, e.g. strong monotonicity; b) exclusion restrictions, which rule out direct causal effects; c) unconfoundedness, which rules out unobserved common causes of variables. The approach produces both sharp sensitivity curves and robustness values, i.e., the smallest sensitivity level at which the sign of a causal effect stops being identified. The applicability of the method is demonstrated by generalizing existing proportion-of-violations approaches, deriving new sensitivity analysis methods for instrumental variable and selection-on-observables designs, and reanalyzing empirical results.

Keywords: sensitivity analysis, causal inference, instrumental variables, partial identification

1 Introduction

In recent decades, the quality of scientific results in the social sciences has improved due to a shift in focus toward better research designs and identification strategies (Angrist and Pischke, 2010). This shift, often referred to as the "credibility revolution," emphasizes transparency in causal assumptions and the use of domain expertise to justify them. For instance, researchers often rely on randomized experiments and natural experiments, in which causal conclusions depend on assumptions that can be more directly connected to the study design. Yet, every causal research design relies on assumptions that cannot be fully verified from observed data alone. In particular, in experiments with noncompliance, identifying effects of treatment received might require assumptions about treatment compliance, whereas instrumental variable designs depend on exclusion restrictions, and observational studies typically require assumptions about the absence of unobserved confounding.

For many assumptions in applied research, though, debates about plausibility involve a false dichotomy: either an assumption is defensible or it is not. Either monotonicity holds or it does not, either exclusion holds or it does not, either treatment assignment is unconfounded or it is not. Sensitivity analysis provides a way to move beyond this binary perspective by evaluating how causal conclusions change under relaxations of these assumptions. For example, one might ask how many defiers or how many units violating the exclusion restriction would be needed to overturn an instrumental variables result, or how much unobserved confounding would be required to overturn a study's conclusion. These methods have developed unevenly across research designs, with sensitivity analyses for selection on observables being far more common than those for other designs. Researchers routinely assess how ro-

bust observational estimates are to violations of unconfoundedness, but comparable tools are less commonly available for assumptions such as monotonicity, exclusion restriction, or other constraints. In this paper, I propose a widely applicable approach that accommodates these assumptions under a single unified framework and allows researchers to easily relax a broad range of additional restrictions beyond unconfoundedness. The method is developed for discrete causal models, although it can be extended to continuous outcomes through discretization and to many continuous covariates through modeling.

The key mathematical observation is simple. For discrete causal models, the proportion of units violating many common assumptions can be written as a polynomial in the model parameters. Bounding this polynomial by a sensitivity parameter therefore turns the relaxation into a partial identification problem in the framework of Duarte et al. (2024). When the resulting program is globally optimized, its extrema are sharp for the specified sensitivity model. The approach then proceeds as follows. A researcher specifies an initial causal model and a relaxed model in which the assumption under evaluation is not imposed. The parameter $\theta \in [0, 1]$ bounds the proportion of the population for which the assumption is relaxed. $\theta = 0$ recovers the initial model and $\theta = 1$ the fully relaxed model. The sensitivity analysis method shows how the identified bounds for the estimand change as θ increases. This allows researchers to report not only how the identified information about a certain causal conclusion changes with relaxations of an assumption but also the sensitivity level at which the sign of the conclusion stops being identified.

More specifically, this paper makes two main contributions. First, Section 4.1 introduces the natural sensitivity analysis framework, which formulates sensitivity analyses through restrictions on generalized principal strata and their probability distributions. In particular,

relaxations of three types of causal assumptions are constructed (Section 4.2). The first type involves violations of functional restrictions, which limit the possible ways that one variable can respond to another. For example, in experiments with noncompliance, the monotonicity or "no defiers" assumption rules out units whose treatment uptake moves opposite to the randomized encouragement. Other examples include linearity and interaction restrictions. The second type involves violations of exclusion restrictions, which rule out direct causal effects. A canonical example is the instrumental variables exclusion restriction, under which a randomized encouragement affects the outcome only through its effect on treatment uptake. The third type involves violations of unconfoundedness assumptions, allowing unobserved common causes that the original model rules out. Because many causal assumptions can be understood as belonging to one of these three classes, the framework provides a common language for comparing sensitivity analyses across a wide range of research designs.

Second, the paper provides an algorithmic procedure for transforming a sensitivity analysis problem into a partial identification problem (Section 4.3). The main step is to provide a causal graphical model indicating which assumption should be relaxed. Given this initial graphical model with the questionable assumption and an estimand, the procedure identifies the corresponding class of violations, augments the model to admit those violations, adds a constraint limiting their prevalence by a parameter θ, and then optimizes the estimand over the resulting model. Sharp sensitivity curves are therefore obtained by solving a sequence of ordinary partial identification problems indexed by θ. This reduction makes it possible to use existing partial identification tools for sensitivity analysis. To implement the method, I construct polynomial programs for optimizing standard estimands in discrete causal models while incorporating observed data and assumptions specified by the user. Once the sen-

sitivity problem has been translated into the corresponding partial identification problem that permits violations, I either obtain numerical bounds by global optimization solvers or employ optimization techniques to derive closed-form solutions. Under the regularity conditions stated in Appendix D, recentered subsampling provides pointwise confidence limits for the sensitivity bounds and a one-sided lower confidence bound for robustness.

To show the applicability of the framework and as major contributions of this paper, in Section 5, I derive new sensitivity analyses for two canonical research designs. First, I propose a sensitivity analysis method in terms of sharp bounds on Local Average Treatment Effect (LATE) under maximum proportions of defiers or violations of the exclusion restriction, extending the results of Richardson and Robins (2010) (Section 5.1). Second, for binary treatment and outcome, I derive sharp bounds on the average treatment effect under a confounding mixture model with a bounded unrestricted component. This generalizes the approach of Bonvini and Kennedy (2022) by removing additional independence restrictions on component membership (Section 5.2). Finally, Section 6 illustrates these results through empirical applications to habit formation in voter turnout (Coppock and Green, 2016) and the effect of education on Brexit voting (Hobolt, 2016).

2 Related Literature

This paper contributes to several strands of work on sensitivity analysis and causal inference. The closest precursor is the literature on sensitivity analysis under selection on observables, where researchers study how omitted confounding would have to behave to overturn a substantive conclusion. In that setting, a wide variety of methods have been proposed to assess

whether conclusions are robust to assumption violations (e.g. Cornfield et al., 1959; Rosenbaum and Rubin, 1983; Imbens, 2003; Tan, 2006; Blackwell, 2014; VanderWeele and Ding, 2017; Masten and Poirier, 2018; Zhao et al., 2019; Cinelli and Hazlett, 2020; Chernozhukov et al., 2026; Dorn and Guo, 2023). The paper also relates to a broader literature that studies sensitivity analysis outside the classic confounding setting, such as the many methods developed for instrumental variable designs (Small, 2007; Cinelli and Hazlett, 2025), mediation (Imai et al., 2010), interference (VanderWeele et al., 2014), external validity (Huang, 2024), and many other specific designs, estimands, and causal models. One goal of the present paper is to replace that piecemeal approach with a unified framework that handles multiple assumption types using a common language.

More specifically, the proposed method is closely connected to a class of approaches that I call *natural sensitivity analyses*: methods that index departures from an assumption by the proportion of units for which it may fail. In assessing sensitivity to unconfoundedness, an early example is Rosenbaum (1987), which allows a specified fraction of matched pairs to be confounded, although the method is developed specifically for exact matching. Recently, Bonvini and Kennedy (2022) use a mixture model with a binary selector where a confounded distribution is selected with probability ϵ and a distribution satisfying unconfoundedness is selected with probability $1-\epsilon$. In their main analysis, they derive sharp bounds on the average treatment effect as a function of ϵ, but, for computational tractability reasons, under the additional assumption that the selection process is, conditionally on the observed covariates, independent of both treatment and outcome. This paper uses its general framework to derive new sharp bounds on the ATE for binary treatment and outcome, without this additional assumption. For instrumental variable models, Richardson and Robins (2010) derive bounds

on average causal effects for specified proportions of defiers while maintaining the exclusion restriction. This paper recovers the corresponding monotonicity bounds within a general optimization framework and obtains sensitivity bounds by allowing the defier proportion to vary up to a specified maximum. Section 5.1 also derives bounds under restrictions on the proportion of units violating the exclusion restriction. More recently, Noack (2021) explicitly developed a sensitivity analysis for violations of monotonicity that requires an additional parameter measuring the discrepancy between the potential-outcome distributions of compliers and defiers. Nevertheless, the sensitivity models underlying these approaches can be derived as special cases of the general approach proposed here.

Finally, this paper relates to automated causal identification (Shpitser and Pearl, 2006; Shpitser et al., 2023; Lee et al., 2020, 2024) and partial identification (Sachs et al., 2023). It builds directly on Duarte et al. (2024) by extending their framework to automate sensitivity analysis.

3 Preliminaries

3.1 Notation

I use uppercase letters for random variables, lowercase letters for their realizations, and $\Pr(\cdot)$ for probability. The potential outcome $Y(a)$ denotes the outcome under an intervention fixing $(A = a)$, and more generally, $Y(a_S)$ denotes the potential outcome when a subset S of variables is intervened upon and set to a_S. I adopt the standard "stable unit treatment value assumption" (Imbens and Rubin, 2015). The framework applies to categorical variables, but

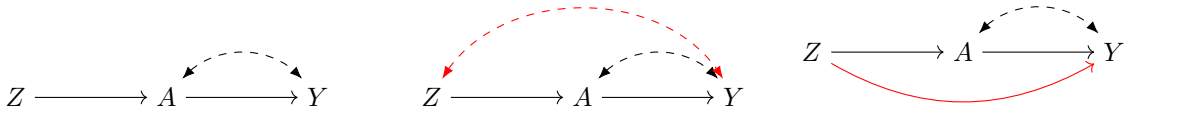
continuous outcomes can be approximated through discretization. For exposition, I use a binary instrumental variable setting, where $Z \in \{z_0, z_1\}$ is an instrument, $A \in \{a_0, a_1\}$ is a treatment, and $Y \in \{y_0, y_1\}$ is the outcome. From causal identification theory, reviewed in Appendix A.1, a model $\mathcal{M}$ represents a set of possible data generating processes (DGPs) over the observed variables $V = V_1, \ldots, V_I$ and the relevant potential outcomes, together with a collection of assumed constraints $\mathcal{C} = C_\lambda : \lambda \in \Lambda$ on these distributions, and an estimand $\mathcal{E}$ represents a quantity of interest as a function of the DGP. The goal of the identification is to characterize all possible values of $\mathcal{E}$ among DGPs of $\mathcal{M}$ that are compatible with an observed data distribution.

Graphs are also used to encode sets of causal assumptions in our analysis. Specifically, I use acyclic directed mixed graphs (ADMGs), which consist of vertices V, directed edges ($E_d \subset V \times V$), and bidirected edges ($E_b \subset V \times V$), with no directed cycles (Richardson et al., 2023). Under a structural causal model associated with the graph, the potential outcome $Y(a_S)$ is the value of Y induced after all arrows into the variables in $S \subset V$ are removed and those variables are externally set to a_S (Pearl, 2009). Figure 1 illustrates an ADMG for the instrumental variable setting. The bidirected edge between A and Y indicates that both treatment and outcome are confounded by unobserved variables. Because Z also has no parents, the absence of bidirected edges between Z and Y and between Z and A represents the assumption of random assignment, while the absence of a direct arrow from Z to Y represents the exclusion restriction. The typical "no defier" assumption has no direct graphical representation and is separately notated as $\Pr(A(z_1) = a_0, A(z_0) = a_1) = 0$.

The sensitivity analysis theory developed here builds on the generalized principal strata framework of Duarte et al. (2024), which parameterizes the joint distribution of potential

outcomes while allowing for unobserved confounding. The procedure called "canonicalization" constructs the generalized principal strata for a model and provides the basis for the global optimization problems used below. This procedure is reviewed in Appendix A.2.

Figure 1: **IV data generating process.** Panel (a) shows the standard IV graph. Panels (b) and (c) add edges that permit violations of random assignment and the exclusion restriction, respectively. Monotonicity is imposed separately as a functional restriction.



(a) Standard IV (b) Randomization violation (c) Exclusion restr. violation

4 Theory for Natural Sensitivity Analyses

In this section, I show that sensitivity analysis of causal assumptions can be formulated as a causal identification problem. Specifically, I study how the identified set for a causal estimand $\mathcal{E}$ changes when some of the constraints in $\mathcal{C}$ are relaxed. A sensitivity analysis problem is therefore defined by a causal model, an observed data distribution, an estimand, and sensitivity parameters that govern the permitted relaxations of the original assumptions.

This formalization has two advantages. First, the same machinery developed for automated point and partial identification can be used to compute solutions to sensitivity analysis problems (Duarte et al., 2024; Sachs et al., 2023). I adopt this perspective below, showing how assumption relaxations can be displayed using causal graphs and interpretable parameters and how the resulting problems can be solved automatically. Second, the formalization provides a common language for comparing sensitivity analysis methods and clarifying how their assumptions and results differ.

4.1 Natural Sensitivity Analysis Problem

The framework assesses sensitivity by changing the proportion of the population for which an assumption is relaxed. The main idea is to report how sharp bounds for an estimand $\mathcal{E}$ vary as the permitted proportion increases. For instance, in the binary IV example, instead of imposing the "no defiers" assumption and obtaining a point identified LATE, one can ask what the identified set would be if defiers made up at most 1%, 5%, or 10% of the population.

A sensitivity analysis begins with a causal model $\mathcal{M}$, an observed data distribution P_V, and an estimand $\mathcal{E}$. Together, the triple $(\mathcal{M}, P_V, \mathcal{E})$ defines the initial causal inference problem. The model $\mathcal{M}$ contains the DGPs over variables $V = \{V_1, \ldots, V_I\}$ that satisfy the assumed constraints $\mathcal{C} = \{C_\lambda : \lambda \in \Lambda\}$. Compatibility with P_V further restricts this model to DGPs that imply the observed data distribution. In the running binary IV example, the constraints include no defiers (C_{ND}), exclusion of a direct effect of the instrument on the outcome (C_{ER}), and random assignment of the instrument (C_{UC}), as described in Section 3.

The next step is to select a subset $\Lambda_R \subseteq \Lambda$ of assumptions to relax. For example, standard sensitivity methods may replace unconfoundedness with restrictions on the association between treatment A and an unobserved confounder U, and between U and the outcome, expressed using risk ratios (VanderWeele and Ding, 2017) or partial R^2 (Cinelli and Hazlett, 2020). The natural sensitivity analysis considered here instead requires, for every $\lambda \in \Lambda_R$, an event $\mathcal{V}_\lambda$ whose probability under DGP P gives the proportion of the population for which C_λ is relaxed. A sensitivity parameter $\theta_\lambda^{\max}$ then places the upper bound $P(\mathcal{V}_\lambda) \leq \theta_\lambda^{\max}$.

Following the terminology introduced in Section 2, I refer to this class of methods as *natural sensitivity analysis*, because the sensitivity parameter has a direct population inter-

pretation: it bounds the proportion of the population for which the assumption is relaxed.

Definition 1 (Natural Sensitivity Problem)**.** *Let* $(\mathcal{M}, P_V, \mathcal{E})$ *be an initial causal inference problem with constraints* $\mathcal{C} = \{C_\lambda : \lambda \in \Lambda\}$*, and let* $\Lambda_R \subseteq \Lambda$ *index the constraints to be relaxed. Let* $\mathcal{M}_R$ *be the corresponding relaxed model, with all remaining constraints retained. Let* $\mathcal{V}_\lambda$ *denote the event identifying the part of the population for which* C_λ *is relaxed. For* $P \in \mathcal{M}_R$*, its probability* $P(\mathcal{V}_\lambda) \in [0,1]$ *is the corresponding population proportion. For a vector of upper bounds* $\Theta^{\max} = \{\theta_\lambda^{\max} : \lambda \in \Lambda_R\}$*, define*

$$\mathcal{M}(\Theta^{\max}) = \{P \in \mathcal{M}_R : P(\mathcal{V}_\lambda) \leq \theta_\lambda^{\max} \text{ for every } \lambda \in \Lambda_R\}.$$

A natural sensitivity problem is the family of causal inference problems, $\left(\mathcal{M}(\Theta^{\max}), P_V, \mathcal{E}\right)$.

Notice that a natural sensitivity problem has corresponding identified sets $\mathcal{I}_\mathcal{E}(\Theta^{\max})$ for each $\Theta^{\max} \in [0,1]^{|\Lambda_R|}$. It characterizes how $\mathcal{I}_\mathcal{E}(\Theta^{\max})$ changes as the allowed violation proportions in $\Theta^{\max}$ vary. The distinction between $P(\mathcal{V}_\lambda)$ and $\theta_\lambda^{\max}$ is relevant. The former is the exact proportion violated, while the latter is the largest proportion allowed by the sensitivity analysis. For example, $\theta_{\text{ND}}^{\max} = 0.05$ allows the proportion of defiers to take any value between 0% and 5%. It does not assert that exactly 5% of the population consists of defiers. Varying $\Theta^{\max}$ traces how the identified set changes as larger violations are admitted. Later, I also show that two or more assumptions can be relaxed simultaneously without changing the method.

The natural sensitivity analysis problem also yields a quantity of particular interest: the *robustness value*. It is the sensitivity level at which the sign of the effect stops being identified.

Definition 2 (Robustness Value)**.** *Let $\mathcal{I}_\lambda(\theta)$ be the identified set obtained from the procedure in Definition 1, and assume $\mathcal{I}_\lambda(0)$ is nonempty. The sign is robust at θ if the identified set is entirely positive or entirely negative. The robustness value $\theta^\star_\lambda$ is the smallest sensitivity level at which this sign conclusion fails, taking the infimum if there is no smallest value. It is zero if the sign is not identified at $\theta = 0$. By convention, it is one both when the first sign failure occurs at $\theta = 1$ and when the sign remains identified for every $\theta \in [0, 1]$.*

4.2 Representing and Interpreting Assumption Violations

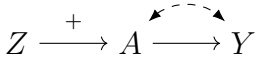
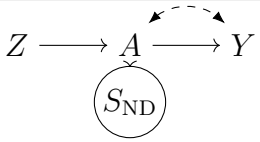
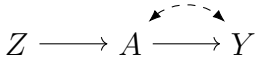
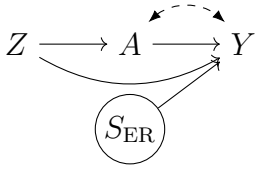
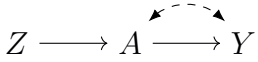
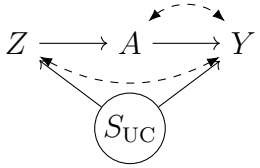
As discussed, natural sensitivity analysis uses parameters that bound the proportion of units violating an assumption. I distinguish three types of assumptions: functional assumptions, exclusion restrictions, and unconfoundedness assumptions. Each has a transparent graphical display. The sensitivity variables introduced below are additional binary nodes in the augmented graph.

First, *functional assumptions* correspond to constraints on how one variable affects another. If a model assumes that encouragement Z causes treatment A but restricts the form of this causal function, this constitutes a functional constraint. The "no defier" assumption discussed above is a unit level functional constraint: among the possible responses of treatment A to instrument Z, it rules out defiers while permitting compliers, never takers, and always takers. Functional constraints can also be stated at the population level. For example, the additive causal mean restriction $\mathbb{E}[Y(z, a)] = \alpha + \beta a + \gamma z$ rules out an interaction in the mean potential outcome surface. Both individual-level and population-level functional restrictions are representable in the polynomial program. Next, consider *exclusion restric-*

tion assumptions, which rule out the presence of a direct causal influence from a variable X to another variable Y that is not mediated by other variables. In graphical language, this is represented by the absence of an $X \to Y$ arrow. Finally, let C collect the observed covariates on which the analysis conditions, and consider unconfoundedness between treatment A and its vector of potential outcomes, $\mathbf{Y}_A = (Y(a) : a \in \mathcal{S}(A))$. Unconfoundedness means $A \perp\!\!\!\perp \mathbf{Y}_A \mid C$. Appendix A.3 gives examples of how these restrictions and their relaxations enter the polynomial program.

To handle assumption relaxations, I introduce a new type of graphical representation, called a *sensitivity graph*, which augments the original causal graph with additional nodes and edges that encode the relaxation of the assumptions and is formalized in Definition 9 in Appendix B.2. For each relaxation, a sensitivity node pointing to the location of the restriction is added, as illustrated in Table 1.

Table 1: **Assumption relaxations in the IV model.** The binary sensitivity nodes indicate where restrictions are relaxed: monotonicity at A, exclusion at Y, and exogeneity through dependence between Z and the outcome response. Their probabilities are bounded by the corresponding sensitivity parameters. The + sign denotes monotonicity in the original model.

Assumption	**Original Model**	**Sensitivity Graph**
Functional		
Exclusion Restriction		
Exogeneity		

4.3 Formalizing the relaxations and the problem

4.3.1 Perturbation and sensitivity functions

As we have defined, $\Lambda_R \subseteq \Lambda$ indexes the relaxed assumptions. Let $K_R = |\Lambda_R|$, and for every $\lambda \in \Lambda_R$ let $\mathcal{V}_\lambda$ be its specified violation event. Let $\Theta = (\theta_\lambda : \lambda \in \Lambda_R) \in [0,1]^{K_R}$ be the vector of realized violation proportions. The causal inference assumptions studied here take the form $P(\mathcal{V}_\lambda) = 0$. For example, the "no defiers" assumption is $P(A(z_0) = a_1, A(z_1) = a_0) = 0$. Replacing 0 with θ_{ND}, we have an exact perturbation, fixing the proportion of defiers. This construction can be repeated for each $\lambda \in \Lambda_R$. In fact, even more complex constraints, such as exclusion restriction and unconfoundedness, can be represented in this way, as will be shown.

For the vector Θ, let $\mathcal{M}_=(\Theta; P_V)$ contain the DGPs that remain compatible with the observed distribution and stated assumptions and satisfy $P(\mathcal{V}_\lambda) = \theta_\lambda$ for every $\lambda \in \Lambda_R$. Solving the partial identification problem associated with this model gives the range of estimand values under the violation proportions. This range defines a perturbation function.

Definition 3 (Perturbation function)**.** *The perturbation function* $f : [0,1]^{K_R} \to 2^{\text{Codomain}(\mathcal{E})}$ *is*

$$f(\Theta) = \{\mathcal{E}(P) : P \in \mathcal{M}_=(\Theta; P_V)\}.$$

For a scalar estimand, write $L(\Theta) = \inf f(\Theta)$ *and* $U(\Theta) = \sup f(\Theta)$ *whenever* $f(\Theta)$ *is nonempty. If no compatible DGP has the exact violation vector* Θ*, then* $f(\Theta) = \varnothing$*.*

The perturbation function evaluates conclusions at specified violation proportions. For

instance, suppose that the other IV assumptions are valid and exactly 20% of units are defiers. Then $\Theta = (0.2)$ and the modified constraint is $P(A(z_0) = a_1, A(z_1) = a_0) = 0.2$. The set $f(0.2)$ gives the admissible LATE values for this exact perturbation.

One challenge with this approach is that researchers often lack precise information about the exact proportion of violations. Consequently, the perturbation function may be most useful for checking how solutions change as Θ varies. Researchers may be more comfortable specifying plausible upper bounds. In the IV example, one can ask how the results change when the proportion of defiers is at most 0.2 rather than exactly 0.2. These cases yield sensitivity analysis functions.

Definition 4 (Sensitivity analysis function)**.** *Let $\Theta^{\max} \in [0,1]^{K_R}$ be a vector of upper bounds for the K_R violation proportions. The sensitivity analysis function $g : [0,1]^{K_R} \to 2^{\text{Codomain}(\mathcal{E})}$ is*

$$g(\Theta^{\max}) = \bigcup_{\Theta \leq \Theta^{\max}} f(\Theta) = \mathcal{I}_{\mathcal{E}}(\Theta^{\max}),$$

where the vector inequality is componentwise. For a scalar estimand, its sharp lower and upper bounds are $\inf_{\Theta \leq \Theta^{\max}} L(\Theta)$ and $\sup_{\Theta \leq \Theta^{\max}} U(\Theta)$ over feasible exact perturbations.

4.3.2 Individual-level relaxations

I first consider individual-level constraints, i.e. those that eliminate a subset of generalized principal strata by setting their probability to 0, as in Definition 5.

Definition 5 (Individual-level constraint)**.** *A constraint C_λ is* individual-level *if it requires $P(\mathcal{V}_\lambda) = 0$, where $\mathcal{V}_\lambda$ is a fixed union of generalized principal stratum events.*

For a natural sensitivity analysis, these constraints can be relaxed by allowing $P(\mathcal{V}_\lambda) \leq$

$\theta_\lambda^{\max}$. The following theorem validates this procedure. It shows that this analysis becomes a polynomial program by changing a constraint in the original causal problem. As the principal strata that violate an assumption are specified, their population proportion is obtained by summing their probabilities. Setting this proportion equal to θ gives the perturbation problem, while bounding it by $\theta^{\max}$ gives the sensitivity problem. Solving the latter globally yields the sharp sensitivity curves. The data constraints, the other assumptions, and the estimand remain unchanged. The proof is in Appendix B.1.

Theorem 1 (Individual-level sensitivity)**.** *Let C_λ be an individual-level constraint as in Definition 5. In the finite canonical model, define $I_\lambda(\boldsymbol{u}) = \mathbb{1}\{\{\boldsymbol{U} = \boldsymbol{u}\} \subseteq \mathcal{V}_\lambda\}$. Let $\boldsymbol{\eta}$ be a vector of all the generalized principal strata probabilities and any auxiliary variables. Let $\mathcal{H}_0(P_V)$ contain the canonicalized parameter vectors that reproduce P_V, satisfy the axioms of probability and all retained restrictions in canonicalization, including auxiliary-variable constraints but excluding C_λ, and leave $\mathcal{E}(\boldsymbol{\eta})$ defined. Then:*

(i) The event probability is the polynomial

$$P_{\boldsymbol{\eta}}(\mathcal{V}_\lambda) = \sum_{\boldsymbol{u} \in \mathcal{S}(\boldsymbol{U})} I_\lambda(\boldsymbol{u}) \prod_{k=1}^{K} p_{k,u_k}. \tag{1}$$

Here $P_{\boldsymbol{\eta}}$ denotes the distribution determined by $\boldsymbol{\eta}$.

(ii) For $\theta, \theta^{\max} \in [0,1]$, define

$$\mathcal{H}_{\lambda,=}(\theta; P_V) = \{\boldsymbol{\eta} \in \mathcal{H}_0(P_V) : P_{\boldsymbol{\eta}}(\mathcal{V}_\lambda) = \theta\},$$

$$\mathcal{H}_{\lambda,\leq}(\theta^{\max}; P_V) = \{\boldsymbol{\eta} \in \mathcal{H}_0(P_V) : P_{\boldsymbol{\eta}}(\mathcal{V}_\lambda) \leq \theta^{\max}\}.$$

These programs represent the exact perturbation problem and the natural sensitivity problem of Definition 1, respectively. Their images under $\mathcal{E}$ are $f(\theta)$ and $g(\theta^{\max})$, with

$$g(\theta^{\max}) = \bigcup_{0 \leq \theta \leq \theta^{\max}} f(\theta).$$

For each feasible problem, the infimum and supremum of the scalar estimand over the corresponding polynomial program give its sharp bounds.

The theorem has some immediate consequences. The first one is that allowing a larger maximum proportion of violations cannot make the identified set smaller.

Corollary 1 (Sensitivity Function Monotonicity)**.** *For any $0 \leq \theta_1^{\max} \leq \theta_2^{\max} \leq 1$,*

$$g(\theta_1^{\max}) \subseteq g(\theta_2^{\max}).$$

Consequently, wherever the sensitivity sets are nonempty, the sharp lower endpoint is nonincreasing and the sharp upper endpoint is nondecreasing:

$$L^{\text{sens}}(\theta_2^{\max}) \leq L^{\text{sens}}(\theta_1^{\max}), \qquad U^{\text{sens}}(\theta_2^{\max}) \geq U^{\text{sens}}(\theta_1^{\max}).$$

More generally, for several sensitivity parameters, $\Theta_1^{\max} \leq \Theta_2^{\max}$ componentwise implies $g(\Theta_1^{\max}) \subseteq g(\Theta_2^{\max})$. This monotonicity does not hold for the exact perturbation function $f(\theta)$, because models with different exact violation proportions are not in general nested.

Another relevant consequence is that this formulation admits a representation of the sensitivity problem as a local mixture model, in which the initial constrained model is chosen

with probability $1-\theta^{\max}$, while the relaxed model, in which the assumption is removed, with probability $\theta^{\max}$. This representation is especially relevant because it suggests some mixture model sensitivity analyses, which were proposed in the literature, are a consequence of this construction.

Corollary 2 (Mixture representation)**.** *Suppose $I_\lambda(\boldsymbol{u})$ depends only on one canonical block U_j. Let $\mathcal{S}_j^{\text{res}}$ contain the strata in which the restriction holds, and suppose this set is nonempty. A principal strata distribution satisfies $P_{\boldsymbol{\eta}}(\mathcal{V}_\lambda) \leq \theta^{\max}$ if and only if it can be written as*

$$P_{U_j} = (1-\theta^{\max})P_{U_j}^{\text{res}} + \theta^{\max} P_{U_j}^{\text{rel}},$$

where $P_{U_j}^{\text{res}}$ is supported on $\mathcal{S}_j^{\text{res}}$ and $P_{U_j}^{\text{rel}}$ is unrestricted on $\mathcal{S}(U_j)$. Substituting the combined principal strata distribution into every original polynomial constraint gives an equivalent parameterization of the same causal model.

The theorem applies only to assumptions expressed as individual-level constraints. This naturally includes exclusion restrictions, as formalized in Lemma 2. The key idea is that an exclusion restriction becomes a functional restriction once the omitted arrow is added: the outcome principal strata must be invariant to the added parent while its other inputs are held fixed. For example, in the binary IV example, the exclusion restriction can be represented either by the graph in Figure 1a, which omits the arrow $Z \to Y$, or by the graph in Figure 1c, which includes that arrow but imposes the individual-level assumption that $P(Y(z_1, a) = y, Y(z_0, a) = y') = 0$ for any a and every pair $y \neq y'$.

4.3.3 Population-level relaxations

In this subsection, I show how the results from the previous subsection extend naturally to population-level assumptions, including unconfoundedness and many functional constraints.

Definition 6 (Population-level constraint)**.** *A constraint C_λ is* population-level *if it requires $h_\lambda(\boldsymbol{\eta}) = 0$, $h_\lambda(\boldsymbol{\eta}) \leq 0$ or $h_\lambda(\boldsymbol{\eta}) \geq 0$, where h_λ is a polynomial function depending on the strata probability $\boldsymbol{\eta}$.*

By this definition, population-level constraints subsume individual-level ones, but do not need to require the probability of a particular stratum to be 0. For example, population-level monotonicity, $\mathbb{E}[A(z_1)] \geq \mathbb{E}[A(z_0)]$, requires the proportion of compliers to be at least as large as that of defiers, while individual-level monotonicity excludes defiers entirely. The population-level restriction can therefore hold even when defiers are present, and its violation cannot in general be assigned to particular units. The current framework, therefore, does not allow a direct extension of the treatment of individual-level to population-level constraints, but this can be circumvented through an additional construction.

To relax a particular population-level constraint, introduce a binary component indicator H. Impose C_λ on the response distribution conditional on $H = 0$, when that component has positive probability, and leave the $H = 1$ component unrestricted by C_λ. The natural sensitivity parameter bounds its weight, $P(H = 1) \leq \theta_\lambda^{\max}$. The data and all retained model restrictions apply to the recovered population distribution. This extends the natural sensitivity framework through an augmented model: $P(H = 1)$ is a component proportion and need not equal the probability of a fixed event violating C_λ. Definition 10 and Lemma 3 in Appendix B.4 give the formal construction and establish invariance under unrestricted

augmentation.

In this sense, the enlargement is only a refinement of the generalized principal strata parameterization, so that solving a problem without imposing any restriction on the auxiliary variable yields the same result as solving the corresponding problem without introducing that variable. Restrictions on H, such as the componentwise independence condition imposed below, are what can change the identified set. An important consequence of this augmentation is that assumptions that were population-level under the original model admit individual-level representation under the augmented one. Therefore, all the results from Theorem 1, including its corollaries, apply directly to population-level constraints as well. This is stated in the following Lemma 1, and the formal proof is in Appendix B.4.

Lemma 1 (Individual-level representation under augmentation)**.** *Assume that C_λ holds within the $H = 0$ component of the augmented model. Then imposing $P(H = 1) = 0$ recovers exactly the original model that satisfies C_λ, by projection onto the original variables. Therefore, relaxing this individual-level restriction to $P(H = 1) \leq \theta_\lambda^{\max}$ defines a sensitivity problem to which Theorem 1 and its corollaries apply under their stated conditions.*

Under the augmented model, unconfoundedness can therefore be represented in terms of a polynomial equality. Let A and Y be finite-valued and write $\mathbf{Y}_A = (Y(a) : a \in \mathcal{S}(A))$. For the application below, take the selected district to contain A and Y. Each generalized principal stratum $a_i\mathbf{y}_{\boldsymbol{r}}$ is thereby divided into two augmented strata, $h_0 a_i \mathbf{y}_{\boldsymbol{r}}$ and $h_1 a_i \mathbf{y}_{\boldsymbol{r}}$. The $H = 0$ cells form the component in which unconfoundedness is imposed, whereas the $H = 1$ cells form an unrestricted component. Let $q_{hi\boldsymbol{r}} = P(h_h a_i \mathbf{y}_{\boldsymbol{r}})$ denote the probabilities of the augmented strata. The original principal-stratum probabilities are recovered by

marginalizing H: $P(a_i \mathbf{y}_{\boldsymbol{r}}) = q_{0i\boldsymbol{r}} + q_{1i\boldsymbol{r}}$. The probability of the unrestricted component is then $P(H = 1) = \sum_{i,\boldsymbol{r}} q_{1i\boldsymbol{r}}$. This assumption is studied in Appendix B.4, which includes its formalization in Lemma 5. For binary and confounded A and Y, the augmented strata are $h_s a_i y_{jk}$ for $s, i, j, k \in \{0, 1\}$. Thus, for example, the original stratum $a_0 y_{01}$ is divided into $h_0 a_0 y_{01}$ and $h_1 a_0 y_{01}$.

As a consequence of both Corollary 2 and Lemma 1, the representation of the sensitivity model as a mixture of two different models where the first one that carries the constraint is chosen with probability $1 - \theta$ and the second one that relaxes that constraint with probability θ extends as well to population-level assumptions. Therefore, the mixture model used by Bonvini and Kennedy (2022), discussed further in Section 5.2, is a special case of this construction.

4.4 Solving the sensitivity analysis problem

To solve the optimization problems arising from natural sensitivity analysis, there are two main computational approaches. The first one generates numerical solutions through branch-and-bound algorithms, employed in packages such as SCIP (Hojny et al., 2025). The algorithm `autobounds` introduced by Duarte et al. (2024) can formulate these problems for numerical optimization and, in fact, the exact perturbation function need not be derived first, as the constraints $P(\mathcal{V}_\lambda) \leq \theta_\lambda^{\max}$ can be included directly in the optimization problem defining $g(\Theta^{\max})$. The numerical sensitivity curve is obtained by repeating this optimization over values of $\Theta^{\max}$. The second approach involves obtaining closed-form solutions for some of those problems. One method uses packages for vertex enumeration, such as cddlib (Fukuda,

2021), also used in `autobounds` to solve causal problems that are linear programs. Finally, closed-form solutions can be derived analytically using standard mathematical techniques.

4.5 Statistical Inference

The estimated sensitivity bounds also carry sampling uncertainty. In particular, the challenges related to doing inference on bounds are related to the fact that those quantities are often not Hadamard differentiable, so there are difficulties in deriving their asymptotic distribution. To address these issues, under the regularity conditions in Appendix D, recentered subsampling provides separate pointwise one-sided confidence limits for the sensitivity bounds and a one-sided lower confidence bound for the robustness value. Details on potential issues and on the validity, consistency, and coverage of the confidence sets as well as sufficient regularity conditions are discussed in Appendix D. Among the conditions, continuity and the existence of a limiting probability distribution are required.

5 Sensitivity Analysis in Standard Scenarios

I next apply the approach to derive sensitivity curves for violations of exclusion and "no defiers" in instrumental variables and unconfoundedness in selection-on-observables. Related results for other assumptions are developed in Duarte (2024) and Cooper et al. (2026).

5.1 Instrumental Variables

Consider the binary instrumental variable setting shown in Figure 1. For clarity, I state the results without covariates. As shown in Appendix A.1, standard assumptions of exogene-

ity, exclusion restriction, and monotonicity ("no defiers") point-identify the LATE (Local Average Treatment Effect), which is denoted by $\mathbb{E}[Y(a_1) - Y(a_0) \mid A(z_1) > A(z_0)]$ (Imbens and Angrist, 1994). The robustness of the LATE to unit-level violation of monotonicity and exclusion can be assessed by the following method. First, perturbation functions are derived in the following two theorems, whose proofs are in Appendix C.2.2.

Theorem 2 (Perturbation bounds on the LATE for monotonicity violations)**.** *Let* $C = \{A(z_1) > A(z_0)\}$ *denote the complier event. Suppose exogeneity and exclusion hold, the observed distribution is compatible with the model and a fixed specified* θ_{ND}*, and* $P(C) = P(A(z_1) > A(z_0)) > 0$*. Then the perturbation bounds for a particular* θ_{ND} *are* $l(\theta_{ND})/P(C) \leq LATE \leq u(\theta_{ND})/P(C)$*, where* $p^k_{ij} = P(Y = y_i, A = a_j \mid Z = z_k)$ *for* $i, j, k \in \{0, 1\}$*. The functions* $l(\theta_{\text{ND}})$ *and* $u(\theta_{\text{ND}})$ *are, respectively,*

$$\max \begin{cases} -p^0_{10}, \\ p^1_{11} - P(y_1|z_0), \\ p^1_{11} + p^0_{00} - 1 + \theta_{ND}, \\ -P(C), \\ p^1_{10} - p^0_{10} - \theta_{ND}, \\ p^0_{01} - p^1_{01} - \theta_{ND}, \\ P(y_0|z_0) - P(y_0|z_1) - \theta_{ND}, \\ -p^1_{01}, \\ p^0_{00} - P(y_0|z_1), \end{cases} , \quad \min \begin{cases} 1 - p^0_{10} - p^1_{01} - \theta_{ND}, \\ p^1_{11}, \\ P(y_0|z_0) - p^1_{01}, \\ P(y_0|z_0) - P(y_0|z_1) + \theta_{ND}, \\ p^0_{00}, \\ P(C), \\ p^0_{00} - p^1_{00} + \theta_{ND}, \\ P(y_1|z_1) - p^0_{10}, \\ p^1_{11} - p^0_{11} + \theta_{\text{ND}}, \end{cases}$$

and $P(C) = P(a_1|z_1) - P(a_1|z_0) + \theta_{ND}$*.*

Next, allow the direct $Z \to Y$ arrow shown in Figure 1c. Let θ_{ER} denote the exact proportion of units for whom changing Z changes $Y(z, a)$ for at least one value of a. With binary variables, this is the total probability of strata satisfying $Y(z_0, a) \neq Y(z_1, a)$ for at least one a.

Theorem 3 (Perturbation bounds on the LATE under exclusion violations)**.** *Here, with exclusion relaxed, write* LATE $= \mathbb{E}[Y(Z, 1) - Y(Z, 0) \mid A(z_1) > A(z_0)]$*, which reduces to the usual LATE under exclusion. Suppose exogeneity and monotonicity hold, the observed distribution is compatible with the model and the specified* θ_{ER}*, and* $P(C) = P(A(z_1) > A(z_0)) > 0$*. Then* LATE *satisfies* $l(\theta_{\text{ER}})/P(C) \leq$ LATE $\leq u(\theta_{\text{ER}})/P(C)$*, using* p^k_{ij} *as above.*

The functions $l(\theta_{\mathrm{ER}})$ *and* $u(\theta_{\mathrm{ER}})$ *are, respectively,*

$$\sum_{k=0}^{1} P(z_k) \max \begin{cases} -P(C), \\ -p^k_{1-k,k}, \\ P(y_1|z_1) - P(y_1|z_0) - \theta_{\mathrm{ER}}, \\ p^0_{01} - p^0_{00} + p^1_{00} - p^1_{01} - \theta_{\mathrm{ER}}, \\ p^1_{10} - p^1_{11} - p^0_{10} + p^0_{11} - \theta_{\mathrm{ER}}, \end{cases} , \sum_{k=0}^{1} P(z_k) \min \begin{cases} P(C), \\ p^k_{kk}, \\ P(y_1|z_1) - P(y_1|z_0) + \theta_{\mathrm{ER}}, \\ p^0_{00} - p^0_{01} - p^1_{00} + p^1_{01} + \theta_{\mathrm{ER}}, \\ p^0_{10} - p^0_{11} + p^1_{11} - p^1_{10} + \theta_{\mathrm{ER}}. \end{cases}$$

with $P(C) = P(a_1|z_1) - P(a_1|z_0)$.

Let $p_{z1} = P(Z = 1)$, $\min_z = \min\{p_{z1}, 1 - p_{z1}\}$, *and* $\max_z = \max\{p_{z1}, 1 - p_{z1}\}$. *Define*

$$a_{p_{z1}} = \begin{cases} p^0_{10}, & p_{z1} \leq \frac{1}{2}, \\ p^1_{01}, & p_{z1} > \frac{1}{2}, \end{cases} \qquad b_{p_{z1}} = \begin{cases} p^0_{00}, & p_{z1} \leq \frac{1}{2}, \\ p^1_{11}, & p_{z1} > \frac{1}{2}. \end{cases}$$

For $\theta = \theta_{\mathrm{ER}}$, *the sharp perturbation bounds are*

$$\begin{aligned} L^{\mathrm{pert}}_{\mathrm{ER}}(\theta) = \frac{1}{P(C)} \max \Big\{ & l(\theta), \, -\max_z P(C) + \min_z(\theta - 1), \\ & -\min_z P(C) - |1 - 2p_{z1}| a_{p_{z1}} + \min_z(\theta - 1) \Big\}, \\ U^{\mathrm{pert}}_{\mathrm{ER}}(\theta) = \frac{1}{P(C)} \min \Big\{ & u(\theta), \, \max_z P(C) + \min_z(1 - \theta), \\ & \min_z P(C) + |1 - 2p_{z1}| b_{p_{z1}} + \min_z(1 - \theta) \Big\}. \end{aligned}$$

The sensitivity curves can be calculated from the perturbations.

Theorem 4 (Natural sensitivity bounds for IV assumption violations)**.** *Consider either relaxation separately, under the maintained assumptions of Theorems 2 and 3, respectively, and suppose its sensitivity model is nonempty. Let* $q^z_a = P(A = a \mid Z = z)$ *and* C *denote the complier event.*

(i) For monotonicity sensitivity, let $\mathcal{D}$ *be the set of feasible values* $d \leq \theta^{\max}_{\mathrm{ND}}$ *with* $P(C) = q^1_1 - q^0_1 + d > 0$. *Let* $\mathcal{B}$ *consist of the endpoints of* $\mathcal{D}$ *and the values of* d *within* $\mathcal{D}$ *at which two nonparallel affine terms in either* $l(d)$ *or* $u(d)$ *intersect. The sharp LATE sensitivity endpoints are*

$$L_{\mathrm{ND}}(\theta^{\max}_{\mathrm{ND}}) = \min_{d \in \mathcal{B}} \frac{l(d)}{q^1_1 - q^0_1 + d}, \qquad U_{\mathrm{ND}}(\theta^{\max}_{\mathrm{ND}}) = \max_{d \in \mathcal{B}} \frac{u(d)}{q^1_1 - q^0_1 + d},$$

using one-sided limits at excluded endpoints.

(ii) For exclusion sensitivity, $P(C) = q_1^1 - q_1^0 > 0$ *is fixed and the sharp sensitivity bounds for* LATE *are*

$$L_{\text{ER}}(\theta_{\text{ER}}^{\max}) = \frac{l(\theta_{\text{ER}}^{\max})}{P(C)}, \qquad U_{\text{ER}}(\theta_{\text{ER}}^{\max}) = \frac{u(\theta_{\text{ER}}^{\max})}{P(C)}.$$

5.2 Classic Confounding

Consider a binary treatment and outcome whose relationship may be confounded by unobserved variables. I derive bounds that allow a limited proportion of the population to belong to a component in which confounding is unrestricted. Theorem 5 extends the mixture approach of Bonvini and Kennedy (2022) by deriving closed-form bounds without the additional restrictions discussed in Section 4.3.3. A proof is given in Appendix C.1.

Theorem 5 (Natural sensitivity bounds under classic confounding)**.** *Let* $\theta = \theta_{\text{UC}} \in [0,1]$ *be the nominal weight of the component in which confounding is permitted, and suppose the mixture is compatible with the observed distribution. Write* $p_{ay} = P(A = a, Y = y)$ *and* $t_{ha} = P(H = h, A = a)$*, as in Appendix C.1. The feasible range of* t_{0a} *is*

$$t_{0a}^- = \max\{0, P(A=a) - \theta\} \le t_{0a} \le \min\{1-\theta, P(A=a)\} = t_{0a}^+.$$

At each fixed $t_{0a} > 0$*, define*

$$L_{P(Y(A=a)=y)}(t_{0a}) = p_{ay} + \max\left\{0, \frac{(1-\theta-t_{0a})(t_{0a} - p_{a,1-y})}{t_{0a}}\right\},$$
$$U_{P(Y(A=a)=y)}(t_{0a}) = \theta + t_{0a} - p_{a,1-y} + \min\left\{1-\theta-t_{0a}, \frac{(1-\theta-t_{0a})p_{ay}}{t_{0a}}\right\}.$$

At $t_{0a} = 0$*, these bounds are* p_{ay} *and* $1 - p_{a,1-y}$*, respectively. The sharp marginal bounds are the minimum of* L *and maximum of* U *over the two endpoints* t_{0a}^-, t_{0a}^+*.*

For the ATE, let $L_a(x) = L_{P(Y(A=a)=1)}(x)$ *and* $U_a(x) = U_{P(Y(A=a)=1)}(x)$*. Both potential outcomes must use the same allocation* $t = t_{01}$*, with* $t_{00} = 1 - \theta - t$*. Define*

$$L_{\text{ATE}}(t) = L_1(t) - U_0(1-\theta-t), \qquad U_{\text{ATE}}(t) = U_1(t) - L_0(1-\theta-t).$$

Let $\mathcal{T} = [t_{01}^-, t_{01}^+]$ *and*

$$\mathcal{B}_L = \{t_{01}^-, t_{01}^+, p_{10}, 1-\theta-p_{01}\} \cap \mathcal{T}, \qquad \mathcal{B}_U = \{t_{01}^-, t_{01}^+, p_{11}, 1-\theta-p_{00}\} \cap \mathcal{T}.$$

The sharp ATE sensitivity bounds are

$$\mathcal{L}_{\mathrm{ATE}}(\theta) = \min_{t \in \mathcal{B}_L} L_{\mathrm{ATE}}(t), \qquad \mathcal{U}_{\mathrm{ATE}}(\theta) = \max_{t \in \mathcal{B}_U} U_{\mathrm{ATE}}(t).$$

The boundary conventions above apply when $t = 0$ *or* $1 - \theta - t = 0$. *The proof of Theorem 5 is given in Appendix C.1.*

6 Empirical Results

6.1 Reanalysis of Coppock and Green (2016)

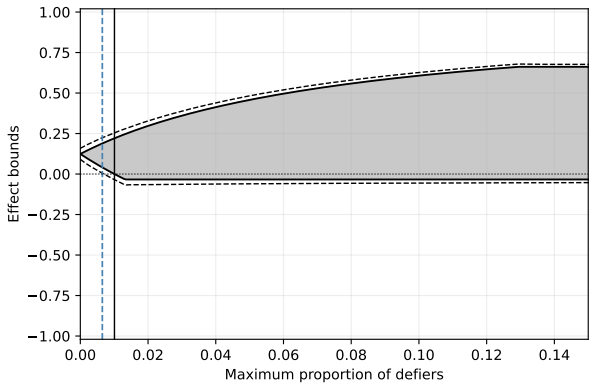
I reanalyze Coppock and Green (2016), who use the social pressure experiment of Gerber et al. (2008) to study whether voting is habit forming. In that experiment, voters were randomly assigned to receive different turnout messages; the strongest treatment informed recipients that their turnout would be monitored and reported to their neighbors. The IV analysis treats the social pressure intervention as encouragement Z, voting in the 2006 primary as treatment A, and voting in a later election as outcome Y. The question is whether voting in election t increases the probability of voting in election $t + 1$.

There are reasons to doubt that monotonicity holds in this setting: given the intrusive nature of the intervention, it is quite possible that social pressure caused some voters not to turn out in election t who otherwise would have. In addition, the exclusion restriction implies that habit formation is the only mechanism through which turnout would increase in election $t+1$: that is, social pressure increases turnout only insofar as it affects turnout in election t, and there is no direct channel through which social pressure affects downstream turnout. However, it is plausible that individuals might have learned from the previous intervention, internalizing the behavior. If individuals have learned or simply remembered being encouraged to vote, then the exclusion restriction is violated.

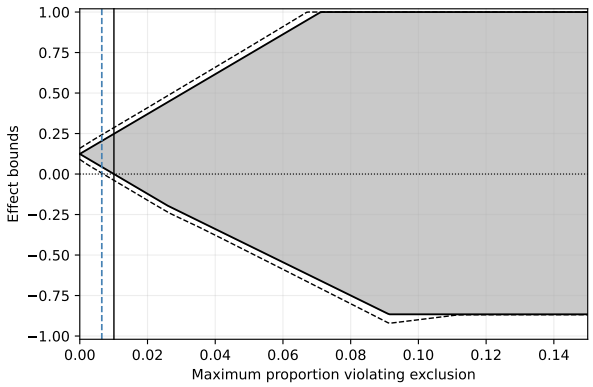
Under monotonicity and exclusion, the estimated LATE is 0.124, with separate 95% one-sided lower and upper confidence limits of 0.091 and 0.159. The analysis uses $n =$ 229,435 observations and 2,000 subsamples of size $m = \lfloor n^{0.8} \rfloor =$ 19,432. Figure 2a relaxes

monotonicity while maintaining exclusion and exogeneity. The estimated robustness value is 0.01008: the sharp lower bound reaches zero when approximately 1.01% of units may be defiers. Its 95% lower confidence bound is 0.0065: the analysis supports robustness to at least 0.65% defiers. This bound addresses possible overstatement of robustness, without seeking an upper confidence bound on the crossing location.

Figure 2: Sensitivity to monotonicity and exclusion violations. Panel (a) bounds LATE; panel (b) bounds the instrument-averaged complier effect. Solid curves are full-sample bounds; dashed curves give separate 95% one-sided confidence limits for the lower and upper bounds. The black vertical line marks the estimated robustness value, 0.01008; the blue dashed vertical line marks its 95% lower confidence bound, 0.0065, in both panels. Both panels use 2,000 subsamples of size 19,432.



(a) Monotonicity: LATE.



(b) Exclusion: instrument-averaged complier effect.

I next relax exclusion by allowing the direct $Z \to Y$ arrow in Figure 1(c). As explained in Section 5.1, ordinary LATE is not defined once exclusion fails, so Figure 2b reports bounds for the instrument averaged complier effect LATE. The lower bound reaches zero at 0.01008, with a 95% lower confidence bound on robustness of 0.0065. These are the sharp analytical sensitivity bounds in Section 5.1. Both IV analyses use the one-sided procedure in Appendix D, with grid spacing 0.0005. When the sensitivity parameter is 0, 14.95% of the subsample draws yielded infeasible problems, but none did at the grid point nearest the estimated robustness value. For infeasible subsamples, I use the bounds $[-1, 1]$. The inference results require the probability of infeasibility to approach zero as the sample size

increases, together with the other conditions in Appendix D. I do not relax exogeneity in this application because the instrument was randomized by design.

6.2 Classic Confounding: Reanalysis of Hobolt (2016)

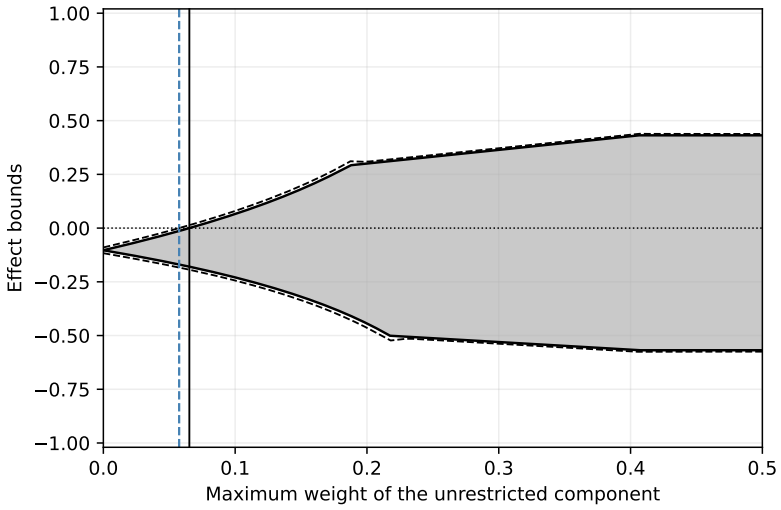
To demonstrate how researchers might apply this method in a case of simple confounding, I replicate and extend Hobolt (2016), an investigation into the individual causes of support for "Brexit," the landmark 2016 referendum in which British voters elected to leave the European Union. The study argues in the abstract that "the divide between winners and losers of globalization was a key driver of the vote". To test this claim, it estimates several multivariate logistic regressions using survey data on support for Brexit. While the study makes numerous causal claims, for illustrative purposes I focus on the following statement: "Going from A-level education to an undergraduate degree reduces the probability of voting Leave by about 10 percentage points, all other things being equal" (1269). In other words, the study claims that voters' level of education was a causal factor influencing the Brexit outcome.

For this reanalysis, I take the estimand to be $\text{ATE}_{3\to4} = \mathbb{E}[Y(d=4) - Y(d=3)]$, the average treatment effect of obtaining an undergraduate degree rather than an A-level education. The original analysis assumes a correctly specified logistic model and conditional unconfoundedness given gender, age, income, and perceived economic conditions (Table 2, Model 1). With or without these controls, the estimated difference in support is approximately -10 percentage points. However, if the included control variables fail to account for all common causes of education and Brexit preference, then the regression contrast generally does not identify the ATE. This might be the case if a variable, such as region or personal connections, affects both obtaining an undergraduate degree and voting against Brexit. If confounding cannot be limited in any way, the resulting bounds are often uninformative.

Using the dataset with 13,229 eligible respondents, the contrast under unconditional unconfoundedness is -0.103, with separate 95% one-sided lower and upper confidence limits

of -0.117 and -0.090. The unrestricted ATE bounds are $[-0.568, 0.432]$. As emphasized in the introduction of this paper, there is no need to restrict the alternative to unconfoundedness or not. Using the approach advocated in this paper, I allow a component in which education may be confounded to have weight at most $\theta_{\text{UC}}^{\max}$. In the remaining component, education is unconfounded. This mixture is a formal decomposition of the population and need not correspond to observed groups. Figure 3 plots the resulting sensitivity curve, optimizing both potential outcomes with a common mixture allocation as described in Appendix C.1. The upper bound reaches zero at $\theta_{\text{UC}}^{\max} = 0.06525$, approximately 6.52%. Its 95% lower confidence bound on robustness is 0.0575, supporting the negative sign through at least 5.75% contamination. This uses 2,000 subsamples of size $m = \lfloor n^{0.8} \rfloor = 1{,}982$ and grid spacing 0.0005. Whether this represents substantial robustness depends on the plausible weight of such a component.

Figure 3: Contamination sensitivity analysis of the ATE. The horizontal axis gives the maximum weight of the component in which confounding is unrestricted, $\theta_{\text{UC}}^{\max}$. Solid curves are full-sample bounds; dashed curves give separate 95% one-sided confidence limits from 2,000 recentered subsamples of size 1,982. The black vertical line marks the estimated robustness value, 6.52%; the blue dashed vertical line marks its 95% lower confidence bound, 5.75%.



7 Conclusion

In this paper, I propose a natural sensitivity analysis approach that applies to a broad class of causal models. The framework represents relaxations of individual-level and population-level constraints as polynomial optimization problems. These problems can be solved numerically using `autobounds`, and closed-form perturbation and sensitivity bounds can be obtained in some cases. The applications examine habit formation in voting and the relationship between education and Brexit preferences. In each case, the robustness value measures the sensitivity level at which the sign of the effect stops being identified. The substantive interpretation of this value remains with the researcher. The resulting bounds are sharp when the optimization reaches the global extrema of the stated model, or when sharpness is established analytically.

Disclosure Statement

The author reports there are no competing interests to declare.

References

Andrews, D. W. K. and S. Han (2009). Invalidity of the bootstrap and the m out of n bootstrap for confidence interval endpoints defined by moment inequalities. *The Econometrics Journal 12*(suppl_1), S172–S199.

Angrist, J. D. and J.-S. Pischke (2010). The credibility revolution in empirical economics: How better research design is taking the con out of econometrics. *Journal of Economic Perspectives 24*(2), 3–30.

Balke, A. and J. Pearl (1997). Bounds on treatment effects from studies with imperfect compliance. *Journal of the American Statistical Association 92*(439), 1171–1176.

Bareinboim, E. and J. Pearl (2016). Causal inference and the data-fusion problem. *Proceedings of the National Academy of Sciences 113*(27), 7345–7352.

Blackwell, M. (2014). A selection bias approach to sensitivity analysis for causal effects. *Political Analysis 22*(2), 169–182.

Bonvini, M. and E. H. Kennedy (2022). Sensitivity analysis via the proportion of unmeasured confounding. *Journal of the American Statistical Association 117*(539), 1540–1550.

Bugni, F. A. (2010). Bootstrap inference in partially identified models defined by moment inequalities: Coverage of the identified set. *Econometrica 78*(2), 735–753.

Chernozhukov, V., C. Cinelli, W. K. Newey, A. Sharma, and V. Syrgkanis (2026). Long story short: Omitted variable bias in causal machine learning. *The Review of Economics and Statistics*. doi: 10.1162/REST.a.1705.

Cinelli, C. and C. Hazlett (2020). Making sense of sensitivity: Extending omitted variable bias. *Journal of the Royal Statistical Society Series B: Statistical Methodology 82*(1), 39–67.

Cinelli, C. and C. Hazlett (2025). An omitted variable bias framework for sensitivity analysis of instrumental variables. *Biometrika 112*(2), asaf004.

Cooper, K. R. D., G. Duarte, L. Keele, D. Knox, and J. Mummolo (2026). Learning from imperfect research designs: Automating causal inference when classic assumptions fail. Working paper.

Coppock, A. and D. P. Green (2016). Is voting habit forming? new evidence from experiments and regression discontinuities. *American Journal of Political Science 60*(4), 1044–1062.

Cornfield, J., W. Haenszel, E. C. Hammond, A. M. Lilienfeld, M. B. Shimkin, and E. L.

Wynder (1959). Smoking and lung cancer: recent evidence and a discussion of some questions. *Journal of the National Cancer Institute 22*(1), 173–203.

Dorn, J. and K. Guo (2023). Sharp sensitivity analysis for inverse propensity weighting via quantile balancing. *Journal of the American Statistical Association 118*(544), 2645–2657.

Duarte, G. (2024). Identification of single-treatment effects in factorial experiments. arXiv:2405.09797v1.

Duarte, G., N. Finkelstein, D. Knox, J. Mummolo, and I. Shpitser (2024). An automated approach to causal inference in discrete settings. *Journal of the American Statistical Association 119*(547), 1778–1793.

Fang, Z. and A. Santos (2019). Inference on directionally differentiable functions. *The Review of Economic Studies 86*(1), 377–412.

Frangakis, C. E. and D. B. Rubin (2002). Principal stratification in causal inference. *Biometrics 58*(1), 21–29.

Fukuda, K. (2021). *cddlib Reference Manual*. ETH Zurich. cddlib version 0.94m.

Gerber, A. S., D. P. Green, and C. W. Larimer (2008, February). Social pressure and voter turnout: Evidence from a large-scale field experiment. *American Political Science Review 102*(1), 33–48.

Hobolt, S. B. (2016). The Brexit vote: a divided nation, a divided continent. *Journal of European Public Policy 23*(9), 1259–1277.

Hojny, C., M. Besançon, K. Bestuzheva, S. Borst, J. Dionísio, J. Ehls, L. Eifler, M. Ghannam, A. Gleixner, A. Göß, et al. (2025). The SCIP optimization suite 10.0. arXiv:2511.18580.

Huang, M. Y. (2024). Sensitivity analysis for the generalization of experimental results. *Journal of the Royal Statistical Society Series A: Statistics in Society 187*(4), 900–918.

Imai, K., L. Keele, and T. Yamamoto (2010). Identification, inference and sensitivity analysis for causal mediation effects. *Statistical Science 25*(1), 51–71.

Imbens, G. W. (2003). Sensitivity to exogeneity assumptions in program evaluation. *American Economic Review 93*(2), 126–132.

Imbens, G. W. and J. D. Angrist (1994). Identification and estimation of local average treatment effects. *Econometrica 62*(2), 467–475.

Imbens, G. W. and C. F. Manski (2004). Confidence intervals for partially identified parameters. *Econometrica 72*(6), 1845–1857.

Imbens, G. W. and D. B. Rubin (2015). *Causal Inference for Statistics, Social, and Biomedical Sciences: An Introduction*. Cambridge University Press.

Lee, J. J. R., A. Ghassami, and I. Shpitser (2024). A general identification algorithm for data fusion problems under systematic selection. In *Proceedings of the Fortieth Conference on Uncertainty in Artificial Intelligence*, Volume 244 of *Proceedings of Machine Learning Research*, pp. 2188–2204. PMLR.

Lee, S., J. D. Correa, and E. Bareinboim (2020). General identifiability with arbitrary surrogate experiments. In *Proceedings of The 35th Uncertainty in Artificial Intelligence Conference*, Volume 115 of *Proceedings of Machine Learning Research*, pp. 389–398. PMLR.

Lundberg, I., R. Johnson, and B. M. Stewart (2021). What is your estimand? defining the target quantity connects statistical evidence to theory. *American Sociological Review 86*(3), 532–565.

Masten, M. A. and A. Poirier (2018). Identification of treatment effects under conditional partial independence. *Econometrica 86*(1), 317–351.

Noack, C. (2021). Sensitivity of LATE estimates to violations of the monotonicity assumption. arXiv:2106.06421v1.

Pearl, J. (2009). *Causality: Models, Reasoning, and Inference* (2 ed.). New York: Cambridge University Press.

Politis, D. N., J. P. Romano, and M. Wolf (2001). On the asymptotic theory of subsampling. *Statistica Sinica 11*(4), 1105–1124.

Richardson, T. S., R. J. Evans, J. M. Robins, and I. Shpitser (2023). Nested Markov properties for acyclic directed mixed graphs. *The Annals of Statistics 51*(1), 334–361.

Richardson, T. S. and J. M. Robins (2010). Analysis of the binary instrumental variable model. In R. Dechter, H. Geffner, and J. Y. Halpern (Eds.), *Heuristics, Probability and Causality: A Tribute to Judea Pearl*, Chapter 25, pp. 415–444. College Publications.

Romano, J. P. and A. M. Shaikh (2010). Inference for the identified set in partially identified econometric models. *Econometrica 78*(1), 169–211.

Rosenbaum, P. R. (1987). Sensitivity analysis for certain permutation inferences in matched observational studies. *Biometrika 74*(1), 13–26.

Rosenbaum, P. R. and D. B. Rubin (1983). Assessing sensitivity to an unobserved binary covariate in an observational study with binary outcome. *Journal of the Royal Statistical Society: Series B (Methodological) 45*(2), 212–218.

Sachs, M. C., G. Jonzon, A. Sjölander, and E. E. Gabriel (2023). A general method for deriving tight symbolic bounds on causal effects. *Journal of Computational and Graphical Statistics 32*(2), 567–576.

Shpitser, I. and J. Pearl (2006). Identification of joint interventional distributions in recursive semi-Markovian causal models. In *Proceedings of AAAI-06*, pp. 1219–1226. AAAI Press.

Shpitser, I., Z. Wood-Doughty, and E. J. Tchetgen Tchetgen (2023). The proximal ID algorithm. *Journal of Machine Learning Research 24*(188), 1–46.

Small, D. S. (2007). Sensitivity analysis for instrumental variables regression with overidentifying restrictions. *Journal of the American Statistical Association 102*(479), 1049–1058.

Tan, Z. (2006). A distributional approach for causal inference using propensity scores. *Journal of the American Statistical Association 101*(476), 1619–1637.

VanderWeele, T. J. and P. Ding (2017). Sensitivity analysis in observational research: introducing the E-value. *Annals of Internal Medicine 167*(4), 268–274.

VanderWeele, T. J., E. J. Tchetgen Tchetgen, and M. E. Halloran (2014). Interference and sensitivity analysis. *Statistical Science 29*(4), 687–706.

Zhao, Q., D. S. Small, and B. B. Bhattacharya (2019). Sensitivity analysis for inverse probability weighting estimators via the percentile bootstrap. *Journal of the Royal Statistical Society: Series B (Statistical Methodology) 81*(4), 735–761.

A Identification Theory and Principal Strata methods

A.1 Identification

Typical identification problems are formalized by first defining a causal model $\mathcal{M}$, which represents a set of possible data generating processes (DGPs) over the observed variables $V = V_1, \ldots, V_I$ and the relevant potential outcomes, together with a collection of assumed constraints $\mathcal{C} = C_\lambda : \lambda \in \Lambda$ on these distributions. These constraints may restrict not only the observed data distribution $\Pr(V)$, but also the joint distribution of potential outcomes $\Pr(V(a_1), \ldots, V(a_J))$ for possible treatment values $a_1, \ldots, a_J$. For example, in the running binary IV setting, the assumed constraints include the no-defiers constraint $C_{\text{ND}} : \Pr(A(z_1) = a_0, A(z_0) = a_1) = 0$, the exclusion restriction constraints $C_{\text{ER},a,z,z'} : \Pr\{Y(z,a) \neq Y(z',a)\} = 0$ for every a, z, z', and the unconfoundedness constraint $C_{\text{UC}} : Z \perp\!\!\!\perp \big(A(z_0), A(z_1), \{Y(z,a) : a \in \{a_0, a_1\}, z \in \{z_0, z_1\}\}\big)$. This condition states that the instrument is jointly independent of all relevant potential treatments and potential outcomes. The constraints here can also be separated by representation type in a pair $(\mathcal{G}, \mathcal{F})$, where $\mathcal{G}$ is a causal ADMG and $\mathcal{F}$ is a set of functional restrictions on the response types allowed by the graph. The choice of using graphs is motivated by syntactic convenience, as they provide a compact representation of the constraints and facilitate computation. The constraints implied by a causal graph will be called here *structural constraints*, while the constraints implied by functional restrictions will be called *functional constraints*. In subsection 4.2, this difference will be further discussed in the context of sensitivity analysis.

Second, the identification problem requires a set of observed data, which is typically represented by the complete or partial observational distribution $\Pr(V)$, as in the standard identification case (Shpitser and Pearl, 2006), or mixing observed distributions over potential outcomes as in the causal data fusion case (Bareinboim and Pearl, 2016). Because the observed data imposes a restriction on the full potential-outcome distribution, it can be understood as a constraint on the model. This point of view will be particularly useful in

the context of the solution for partial identification problems, which will require a constrained optimization approach, as discussed below. It is also worth noting that the observed data here is a population distribution, which will be assumed to be known. In practice, the observed data is estimated from a finite sample, which introduces sampling uncertainty, and will be discussed separately in Appendix D.

The third relevant part of the identification problem is the estimand $\mathcal{E}$, which represents the causal quantity of interest. This estimand is typically a computable function of the distribution of potential outcomes $\Pr(V(a_1), \ldots, V(a_J))$. For example, in the running binary IV example, the estimand of interest is the local average treatment effect (LATE) among compliers, which can be expressed as $\mathcal{E} := \mathbb{E}(Y(a_1) - Y(a_0) \mid A(z_1) = a_1, A(z_0) = a_0)$. The identification problem is then to determine whether $\mathcal{E}$ can be uniquely computed from the model, given the observed data.

Definition 7 (Point Identification)**.** *Given a model $\mathcal{M}$, an observed data distribution, and an estimand $\mathcal{E}$, the problem of deciding whether every DGP in $\mathcal{M}$ that is consistent with the observed data distribution yields the same value of $\mathcal{E}$ is called the* point identification problem.

A few points about point identification are worth noting. First, the point identification problem is a purely theoretical problem that involves population quantities, not statistical inference. Second, the point identification problem is a yes-or-no question: either the estimand $\mathcal{E}$ can be uniquely determined from the model and observed data, or it cannot. For that reason, proofs of non-identifiability often involve constructing two DGPs that are consistent with the model and observed data, but yield different values of $\mathcal{E}$. Finally, causal inference practice focuses heavily on obtaining identification results, sometimes to the point of changing an estimand to make it identifiable. For instance, in the IV setting, the LATE is often used instead of the ATE because it is identifiable under the standard IV assumptions with monotonicity while the latter is not. However, the estimand should follow directly from the substantive theory and should not be chosen just for the sake of identifiability, a relevant

point brought up by Lundberg et al. (2021).

Nonetheless, even if the identification problem yields a negative answer, useful results can still be obtained. In particular, one can determine which values of the estimand $\mathcal{E}$ do not contradict the model and observed data. In other words, if a unique value for $\mathcal{E}$ cannot be determined, it is still possible to compute the range of values of $\mathcal{E}$ that satisfy the assumptions and constraints. This is the problem of partial identification, which generalizes point identification by allowing the identified set to contain more than one value.

Definition 8 (Partial identification)**.** *Given a model $\mathcal{M}$, an observed data distribution, and an estimand $\mathcal{E}$, the problem of characterizing the set of values of $\mathcal{E}$ yielded by all DGPs in $\mathcal{M}$ that are consistent with the observed data distribution is called the* partial identification problem. *This set of values is called the* identified set*.*

In simple terms, the partial identification problem asks which values of the estimand $\mathcal{E}$ are compatible with the observed data, given the constraints $\mathcal{C}$ of the model $\mathcal{M}$. If only one value is allowed, then the estimand is point identified; if more than one value is allowed, then the estimand is partially identified. Thus, partial identification generalizes point identification. As an example, in the running binary IV example, the LATE is point identified under the standard IV assumptions including monotonicity (Imbens and Angrist, 1994), but the ATE is only partially identified (Balke and Pearl, 1997). Typically, the identified set is summarized by lower and upper bounds, which delimit the range of values in which the identified set must lie. When the identified set is a closed, bounded interval, it can be written as $[\mathcal{E}_{\min}, \mathcal{E}_{\max}]$. In particular, a researcher would in general be interested in bounds that are *sharp*, meaning that they are the tightest possible bounds for the estimand $\mathcal{E}$ given the model and observed data and are the most informative for the researcher. General algorithms for computing those bounds have been developed in Sachs et al. (2023) and Duarte et al. (2024). It is worth noting that the identified set might not be represented by a closed interval, for instance, if the identified set is a union of disjoint intervals. For a bounded and closed identified set, the lower and upper bounds are respectively its minimum and maximum. Those bounds are

sharp as long as they cannot be improved, i.e. there are no other greater lower and smaller upper bounds that cover the identified set.

In the following sections, I will formalize a general sensitivity analysis framework that builds on the above identification and partial identification problems. The key idea is to study how the identified set for $\mathcal{E}$ changes as I relax some of the constraints in $\mathcal{C}$, allowing for a proportion of units to violate those assumptions. This method allows the assessment of the robustness of causal conclusions to violations of key assumptions.

A.2 A Brief Primer on Generalized Principal Stratification and Autobounds

Manually deriving sharp bounds on effects can be analytically complex and time-consuming. To simplify this process, Duarte et al. (2024) offers a user-friendly algorithm that computes sharp bounds automatically based on the parameters of a causal question, called *autobounds*. I will utilize this algorithm for the sensitivity analyses. For a comprehensive description of the algorithm, refer to Duarte et al. (2024). The algorithm takes four inputs: a causal theory, represented in an ADMG; any additional functional assumption not captured by the ADMG (e.g. "no defiers" or "no interaction"); a causal estimand (e.g. the ATE); and empirical distributions over a set of discrete variables. The algorithm then constructs a polynomial program whose feasible points represent causal models consistent with these inputs. Global minimization and maximization yield the sharp endpoints. If computation stops earlier, the algorithm's certified outer bounds remain valid, although they need not yet be sharp.

Central to this procedure is the idea of principal stratification (Balke and Pearl, 1997; Frangakis and Rubin, 2002), which involves categorizing units in a study based on their potential responses under different conditions. For instance, in the classic IV framework, units are divided into four principal strata with respect to the treatment variable: compliers, defiers, always-takers, and never-takers. The key idea is that any discrete causal model can be represented in terms of principal strata based on how each variable might respond to its

causal past. As the complexity of causal systems increases, the number of principal strata grows rapidly. However, as long as all variables are discrete, the number of strata remains finite, because there are only a limited number of ways discrete variables can interact. Duarte et al. (2024) generalizes this procedure into *generalized principal stratification* to account for unobserved confounders that may be high-dimensional and continuous. When the observed variables are discrete, the number of response types remains finite, while their probabilities, and hence the causal estimand, may vary continuously. This reduction of the causal graph is called *canonicalization*.

In formal terms, the method of generalized principal stratification involves considering, for every variable V_i in a graph, all possible functions that map all possible values of all proximal causes (parents in the graph) to possible values of V_i. If V_i can take on c values and its parents have d possible unique combinations of values, then its principal strata space has cardinality c^d. For instance, in the instrumental variable scenario with binary variables, Z has no observable parent, so its principal strata space is $\{z_0, z_1\}$, representing control and encouraged units. A has only one observable parent, Z, so the strata of A must account for all possible functions from Z to A, which can be compactly written as $\{a_{00}, a_{01}, a_{10}, a_{11}\}$, representing respectively never-takers, compliers, defiers, and always-takers. In this notation, the strata are indexed by binary variables, where the first digit indicates the value of A if Z were 0, and the second digit, if Z were 1. Likewise, for Y, the strata are $\{y_{00}, y_{01}, y_{10}, y_{11}\}$, as Y has only A as an observable parent. After the response types are enumerated, it is still necessary to account for unobserved common causes. In the canonical parameterization of the standard IV model, one disturbance U_Z generates the response type of Z, and a second disturbance U_{AY} generates the combined response type of A and Y. A realization of U_{AY} then defines one joint stratum: $a_{rs}y_{tu}$. In summary, the canonical parameterization of the standard IV model has produced two sets of principal strata: $\{z_0, z_1\}$ for Z and $\{a_{00}y_{00}, \cdots, a_{11}y_{11}\}$ for the joint response type of A and Y, which capture not only the potential outcomes of A and Y, but also their dependence structure. As U_Z and U_{AY} are independent, the joint

distribution of the principal strata is $P(z_i, a_{rs}y_{tu}) = P(z_i)P(a_{rs}y_{tu})$.

The representation with two disturbances is exact for the standard IV model. In more general graphs, canonicalization must retain the structure of the independent disturbances.

Let $V = (V_1, \ldots, V_J)$ be the finite observed variables in a canonical structural causal model and let $U = (U_1, \ldots, U_K)$ be its exogenous disturbances. Canonical disturbances are roots and are mutually independent. Moreover, no disturbance has a set of observed children contained in the child set of another disturbance. The canonicalization algorithm and the finite support construction are given by Duarte et al. (2024).

Each observed variable obeys an equation

$$V_j = f_j\{\mathrm{pa}_V(V_j), \mathrm{pa}_U(V_j)\}.$$

Fixing the disturbances turns this equation into a deterministic response function of the observed parents.

By Proposition 1 of Duarte et al. (2024), a finite categorical representation of the disturbances preserves the full data law. Here, $\mathcal{S}(U_k)$ is the state space of the categorical U_k, and an event $\{U_k = u_k\}$ corresponds to a generalized principal stratum. A category may encode response functions for more than one observed variable. Let

$$P_{U_k} = \{p_{k,u_k} = P(U_k = u_k) : u_k \in \mathcal{S}(U_k)\}, \qquad P_U = (P_{U_1}, \ldots, P_{U_K}).$$

The entries of P_U are the probabilities of the generalized principal strata and are the variables used in the polynomial program. For each k, $p_{k,u_k} \geq 0$ and $\sum_{u_k \in \mathcal{S}(U_k)} p_{k,u_k} = 1$. For $\boldsymbol{u} = (u_1, \ldots, u_K)$, independence gives

$$P(\boldsymbol{U} = \boldsymbol{u}) = \prod_{k=1}^{K} p_{k,u_k}.$$

Fixing $\boldsymbol{u}$ determines a vector of strata, denoted $\rho(\boldsymbol{u})$. Hence, for any event E defined by a

finite conjunction or union of counterfactual statements,

$$P(E) = \sum_{\boldsymbol{u} \in \mathcal{S}(U_1) \times \cdots \times \mathcal{S}(U_K)} \mathbb{1}\{\rho(\boldsymbol{u}) \in E\} \prod_{k=1}^{K} p_{k,u_k}. \tag{2}$$

This is the polynomial representation established by Proposition 2 of Duarte et al. (2024). Observed data restrictions, other assumptions, and causal estimands can therefore be represented by polynomial equalities, inequalities, and polynomial fractions in the canonical probabilities. The exact construction retains all these restrictions; global optimization of the complete program gives sharp bounds relative to the stated model. Duarte et al. (2024) provides the complete canonicalization algorithm. A complete stratification of an instrumental variable model, including the representations of estimands and observed data, can be found in Appendix C.2.1. The idea behind this procedure is that we can maximize or minimize the estimand given data and assumptions (constraints) over the causal model strata (parameters).

Employing stratification, `autobounds` efficiently identifies all principal strata suggested by a causal model and its assumptions. It crucially discards strata that cannot exist given the specified assumptions and observed data, or those that provide redundant information. The program then represents the causal estimand as a polynomial expression based on these principal strata, which can be optimized computationally, subject to the constraints of the causal theory, assumptions, and data. In this form, finding sharp bounds becomes a computational problem. The main challenge is achieving solutions in reasonable time, especially as the number of principal strata grows quickly with the complexity of the variables.

A.3 Examples of assumption restrictions

Mean additivity and individual-level no interaction. For binary A, Z, let r index a generalized principal stratum with probability p_r, and define

$$\Delta_r = Y_r(1,1) - Y_r(0,1) - Y_r(1,0) + Y_r(0,0).$$

Mean additivity imposes $\sum_r p_r \Delta_r = 0$, allowing cancellation across generalized principal strata. Individual-level no interaction instead imposes $p_r = 0$ whenever $\Delta_r \neq 0$.

Exclusion restrictions. To relax an exclusion restriction forbidding a direct $X \to Y$ effect, allow this arrow and expand the generalized principal strata space $\mathcal{R}_Y$ to include strata in which Y changes with X while the other observed parents of Y are held fixed. For example, suppose X and Y are binary and Y has no other observed parents. Without the arrow, Y has the two constant strata $\mathcal{R}_Y = \{y_0, y_1\}$. Allowing the arrow produces $\mathcal{R}_Y = \{y_{00}, y_{01}, y_{10}, y_{11}\}$, where the indices record $Y(x_0)$ and $Y(x_1)$. The exclusion restriction is violated in strata y_{01} and y_{10}. An exact perturbation therefore imposes

$$P(y_{01}) + P(y_{10}) = \theta_{\text{ER}},$$

whereas a natural sensitivity problem imposes

$$P(y_{01}) + P(y_{10}) \leq \theta_{\text{ER}}^{\max}.$$

At zero, with all other model restrictions retained, the expanded model is observationally and causally equivalent to the model without the arrow.

Unconfoundedness. Let C denote observed covariates and $\mathbf{Y}_A = (Y(a) : a \in \mathcal{S}(A))$ the potential-outcome vector. For sensitivity analysis, introduce a component indicator H and impose $A \perp\!\!\!\perp \mathbf{Y}_A \mid C, H = 0$ wherever this conditional distribution is defined. The $H = 1$

component allows arbitrary dependence, and its probability is bounded by the sensitivity parameter. The polynomial factorization also covers a component of zero mass. Lemma 5 and its proof in Appendix B.4 establish the factorization and mixture representation.

B General Sensitivity Theory

B.1 Proof for individual-level sensitivity analysis

Recall Definition 5: a constraint C_λ is individual-level if it requires $P(\mathcal{V}_\lambda) = 0$, where $\mathcal{V}_\lambda$ is a fixed union of generalized principal stratum events. These are the strata that violate the restriction. Let I_λ indicate membership in $\mathcal{V}_\lambda$. Relaxing the restriction allows their total probability to equal θ for perturbation analysis, or to be at most $\theta^{\max}$ for sensitivity analysis.

Theorem 1 (Individual-level sensitivity)**.** *Let C_λ be an individual-level constraint as in Definition 5. In the finite canonical model, define $I_\lambda(\boldsymbol{u}) = \mathbb{1}\{\{\boldsymbol{U} = \boldsymbol{u}\} \subseteq \mathcal{V}_\lambda\}$. Let $\boldsymbol{\eta}$ be a vector of all the generalized principal strata probabilities and any auxiliary variables. Let $\mathcal{H}_0(P_V)$ contain the canonicalized parameter vectors that reproduce P_V, satisfy the axioms of probability and all retained restrictions in canonicalization, including auxiliary-variable constraints but excluding C_λ, and leave $\mathcal{E}(\boldsymbol{\eta})$ defined.*

Then:

(i) The event probability is the polynomial

$$P_{\boldsymbol{\eta}}(\mathcal{V}_\lambda) = \sum_{\boldsymbol{u} \in \mathcal{S}(\boldsymbol{U})} I_\lambda(\boldsymbol{u}) \prod_{k=1}^{K} p_{k,u_k}.$$

Here $P_{\boldsymbol{\eta}}$ denotes the distribution determined by $\boldsymbol{\eta}$.

(ii) For $\theta, \theta^{\max} \in [0,1]$, define

$$\mathcal{H}_{\lambda,=}(\theta; P_V) = \{\boldsymbol{\eta} \in \mathcal{H}_0(P_V) : P_{\boldsymbol{\eta}}(\mathcal{V}_\lambda) = \theta\},$$
$$\mathcal{H}_{\lambda,\leq}(\theta^{\max}; P_V) = \{\boldsymbol{\eta} \in \mathcal{H}_0(P_V) : P_{\boldsymbol{\eta}}(\mathcal{V}_\lambda) \leq \theta^{\max}\}.$$

These programs represent the exact perturbation problem and the natural sensitivity problem of Definition 1, respectively. Their images under $\mathcal{E}$ are $f(\theta)$ and $g(\theta^{\max})$, with

$$g(\theta^{\max}) = \bigcup_{0 \leq \theta \leq \theta^{\max}} f(\theta).$$

For a scalar estimand, the sharp endpoints satisfy

$$L^{\text{sens}}(\theta^{\max}) = \inf_{0 \leq \theta \leq \theta^{\max}} L(\theta), \qquad U^{\text{sens}}(\theta^{\max}) = \sup_{0 \leq \theta \leq \theta^{\max}} U(\theta),$$

omitting infeasible values of θ.

For each feasible problem, the infimum and supremum of the scalar estimand over the corresponding polynomial program give its sharp bounds.

Proof of Theorem 1. The event $\mathcal{V}_\lambda$ is the disjoint union of the canonical cells $\{\boldsymbol{U} = \boldsymbol{u}\}$ for which $I_\lambda(\boldsymbol{u}) = 1$. The canonical blocks are independent, so its probability is

$$\sum_{\boldsymbol{u} \in \mathcal{S}(\boldsymbol{U})} I_\lambda(\boldsymbol{u}) \prod_{k=1}^{K} p_{k,u_k} = P_{\boldsymbol{\eta}}(\mathcal{V}_\lambda).$$

Each summand is a monomial in the principal stratum sizes, so $P_{\boldsymbol{\eta}}(\mathcal{V}_\lambda)$ is a polynomial. This is also an immediate application of Proposition 2 of Duarte et al. (2024) to the event $\mathcal{V}_\lambda$. Any auxiliary variables remain subject to their original defining constraints and do not enter $P_{\boldsymbol{\eta}}(\mathcal{V}_\lambda)$. This proves claim (i).

For any $\boldsymbol{\eta} \in \mathcal{H}_{\lambda,\leq}(\theta^{\max}; P_V)$, set $t = P_{\boldsymbol{\eta}}(\mathcal{V}_\lambda)$. Then $0 \leq t \leq \theta^{\max}$ and $\boldsymbol{\eta} \in \mathcal{H}_{\lambda,=}(t; P_V)$.

This proves one inclusion in

$$\mathcal{H}_{\lambda,\leq}(\theta^{\max}; P_V) = \bigcup_{0 \leq t \leq \theta^{\max}} \mathcal{H}_{\lambda,=}(t; P_V).$$

On the other hand, if $\boldsymbol{\eta} \in \mathcal{H}_{\lambda,=}(t; P_V)$ for some $t \leq \theta^{\max}$, then $P_{\boldsymbol{\eta}}(\mathcal{V}_\lambda) = t \leq \theta^{\max}$, proving the reverse inclusion. Applying $\mathcal{E}$ gives the union of identified sets. Taking infima and suprema gives the envelope formulas.

The construction in terms of generalized principal strata represents the observed data, counterfactual restrictions, and estimand by polynomials or polynomial fractions in the generalized principal stratum sizes. The displayed formula adds one polynomial equality or inequality. Theorem 1 of Duarte et al. (2024), therefore, applies to this polynomial program, so whenever its global extrema are obtained, they are the sharp extrema over precisely the stated model. Proposition 1 of the same paper allows the model to be represented by finitely many generalized principal strata while preserving the joint distribution of observed variables and potential outcomes. When $\mathcal{V}_\lambda$ is defined by the potential outcomes, preserving their joint distribution also preserves $P_{\boldsymbol{\eta}}(\mathcal{V}_\lambda)$. This proves claim (ii).

□

Corollary 3 (Simultaneous individual-level relaxations)**.** *Under the conditions of Theorem 1, consider finitely many individual-level constraints on the same canonical model. Retain all other model constraints and impose one polynomial inequality per restriction:*

$$P_{\boldsymbol{\eta}}(\mathcal{V}_{\lambda_j}) \leq \theta_j^{\max}, \qquad j = 1, \ldots, J.$$

The union and sharp endpoint formulas in that theorem hold with componentwise bounds on Θ*. The violation events may overlap, and independence among them is not required.*

Proof. For a finite collection of assumptions, each $P_{\boldsymbol{\eta}}(\mathcal{V}_{\lambda_j})$ has the same polynomial form. If $\boldsymbol{\eta}$ satisfies $P_{\boldsymbol{\eta}}(\mathcal{V}_{\lambda_j}) \leq \theta_j^{\max}$ for every j, set $\Theta = (P_{\boldsymbol{\eta}}(\mathcal{V}_{\lambda_j}))_{j=1}^J$. Then, $0 \leq \Theta \leq \Theta^{\max}$ and

$\boldsymbol{\eta}$ belongs to the corresponding exact vector set. If each violation proportion is at most its bound, all the sensitivity constraints are satisfied by the model. The strata that violate more than one restriction contribute to the violation proportion of each. Taking the image under the estimand and then its infimum and supremum gives the union and sharp endpoint formulas. □

Corollary 1 (Sensitivity Function Monotonicity)**.** *For any* $0 \leq \theta_1^{\max} \leq \theta_2^{\max} \leq 1$,

$$g(\theta_1^{\max}) \subseteq g(\theta_2^{\max}).$$

Consequently, wherever the sensitivity sets are nonempty, the sharp lower endpoint is nonincreasing and the sharp upper endpoint is nondecreasing:

$$L^{\text{sens}}(\theta_2^{\max}) \leq L^{\text{sens}}(\theta_1^{\max}), \qquad U^{\text{sens}}(\theta_2^{\max}) \geq U^{\text{sens}}(\theta_1^{\max}).$$

More generally, for several sensitivity parameters, $\Theta_1^{\max} \leq \Theta_2^{\max}$ *componentwise implies* $g(\Theta_1^{\max}) \subseteq g(\Theta_2^{\max})$*. This monotonicity does not hold for the exact perturbation function* $f(\theta)$*, because models with different exact violation proportions are not in general nested.*

Proof of Corollary 1. If $\theta_1^{\max} \leq \theta_2^{\max}$, then every $\boldsymbol{\eta}$ satisfying $P_{\boldsymbol{\eta}}(\mathcal{V}_\lambda) \leq \theta_1^{\max}$ also satisfies $P_{\boldsymbol{\eta}}(\mathcal{V}_\lambda) \leq \theta_2^{\max}$. Therefore,

$$\mathcal{H}_{\lambda,\leq}(\theta_1^{\max}; P_V) \subseteq \mathcal{H}_{\lambda,\leq}(\theta_2^{\max}; P_V).$$

Considering the compatible values of $\mathcal{E}$ under each constrained problem proves $g(\theta_1^{\max}) \subseteq g(\theta_2^{\max})$. Therefore, its suprema and infima satisfy the endpoint inequalities. The same argument applied to every component gives the vector result. On the other hand, $P_{\boldsymbol{\eta}}(\mathcal{V}_\lambda) = \theta_1$ does not imply $P_{\boldsymbol{\eta}}(\mathcal{V}_\lambda) = \theta_2$ for $\theta_1 \neq \theta_2$, so the exact perturbation sets need not be nested. □

Corollary 2 (Mixture representation)**.** *Suppose* $I_\lambda(\boldsymbol{u})$ *depends only on one canonical block* U_j*. Let* $\mathcal{S}_j^{\text{res}}$ *contain the strata in which the restriction holds, and suppose this set is*

nonempty. A principal strata distribution satisfies $P_{\boldsymbol{\eta}}(\mathcal{V}_\lambda) \leq \theta^{\max}$ *if and only if it can be written as*

$$P_{U_j} = (1-\theta^{\max})P_{U_j}^{\text{res}} + \theta^{\max} P_{U_j}^{\text{rel}},$$

where $P_{U_j}^{\text{res}}$ *is supported on* $\mathcal{S}_j^{\text{res}}$ *and* $P_{U_j}^{\text{rel}}$ *is unrestricted on* $\mathcal{S}(U_j)$*. Substituting the combined principal strata distribution into every original polynomial constraint gives an equivalent parameterization of the same causal model.*

Proof of Corollary 2. Let $t = P_{\boldsymbol{\eta}}(\mathcal{V}_\lambda) \leq \theta^{\max}$. Since the violation depends only on U_j, the probability outside $\mathcal{S}_j^{\text{res}}$ is t and the probability inside it is $1-t$. For $0 < \theta^{\max} < 1$, define

$$P_{U_j}^{\text{res}}(u_j) = \begin{cases} p_{j,u_j}/(1-t), & u_j \in \mathcal{S}_j^{\text{res}}, \\ 0, & u_j \notin \mathcal{S}_j^{\text{res}}, \end{cases}$$

and

$$P_{U_j}^{\text{rel}}(u_j) = \begin{cases} \dfrac{\theta^{\max}-t}{\theta^{\max}(1-t)} p_{j,u_j}, & u_j \in \mathcal{S}_j^{\text{res}}, \\ \dfrac{p_{j,u_j}}{\theta^{\max}}, & u_j \notin \mathcal{S}_j^{\text{res}}. \end{cases}$$

Both are probability distributions, $P_{U_j}^{\text{res}}$ is supported on the restricted strata, and direct substitution gives

$$P_{U_j} = (1-\theta^{\max})P_{U_j}^{\text{res}} + \theta^{\max} P_{U_j}^{\text{rel}}.$$

If $\theta^{\max} = 0$, take $P_{U_j}^{\text{res}} = P_{U_j}$ and choose $P_{U_j}^{\text{rel}}$ arbitrarily. If $\theta^{\max} = 1$, take $P_{U_j}^{\text{rel}} = P_{U_j}$ and choose $P_{U_j}^{\text{res}}$ arbitrarily on $\mathcal{S}_j^{\text{res}}$.

Conversely, suppose the displayed mixture holds. Since $P_{U_j}^{\text{res}}$ assigns no probability to a

violation,

$$P_{\boldsymbol{\eta}}(\mathcal{V}_\lambda) = \theta^{\max} \sum_{u_j \notin \mathcal{S}_j^{\text{res}}} P_{U_j}^{\text{rel}}(u_j) \leq \theta^{\max}.$$

Only one generalized principal strata distribution has been reparameterized. The other independent blocks remain unchanged. Substituting P_{U_j} into every original expression reproduces the same $\prod_k p_{k,u_k}$. Therefore, the two parameterizations have the same feasible full data laws and the same estimand values. □

Lemma 2 (Exclusion restriction as individual-level constraint)**.** *Suppose the original graph excludes the arrow $X \to Y$, and form the relaxed graph by adding it. Let W collect the other observed parents of Y in the relaxed graph. Within a generalized principal stratum, the exclusion restriction is*

$$Y(x, w) = Y(x', w) \quad \textit{for every } x, x', w.$$

Thus, the exclusion restriction is an individual-level restriction in the relaxed graph. Taking λ in Theorem 1 to be this restriction, I_λ identifies the principal strata for which $Y(x, w) \neq Y(x', w)$ for some x, x', w, and $P_{\boldsymbol{\eta}}(\mathcal{V}_\lambda)$ is their population proportion. Consequently, all the conclusions of Theorem 1 apply.

B.2 Sensitivity Graphs

A sensitivity graph locates the assumptions being relaxed, as defined below. The graph and the stated constraints determine the sensitivity model. A sensitivity model consists of a sensitivity graph together with the support restrictions and any additional constraints on the component and recovered probability distributions.

Definition 9 (Sensitivity graph)**.** *Let $(\mathcal{G}_R, \mathcal{F}_R)$ represent the relaxed model on the original variables V, and let Λ_R index its designated relaxations. For each $\lambda \in \Lambda_R$, specify a binary*

sensitivity indicator S_λ*, a bound* $P(S_\lambda = 1) \leq \theta_\lambda^{\max}$*, and a location* $\chi_\lambda \subseteq V$*. The location contains the variables whose response functions are constrained by a functional restriction, the child* Y *for exclusion of* $X \to Y$*, or the treatment and outcome variables involved in the unconfoundedness restriction. The sensitivity graph adds the vertices* S_{Λ_R} *to* $\mathcal{G}_R$ *and the edges*

$$E_{\text{loc}} = \{S_\lambda \to V_j : V_j \in \chi_\lambda,\ \lambda \in \Lambda_R\}.$$

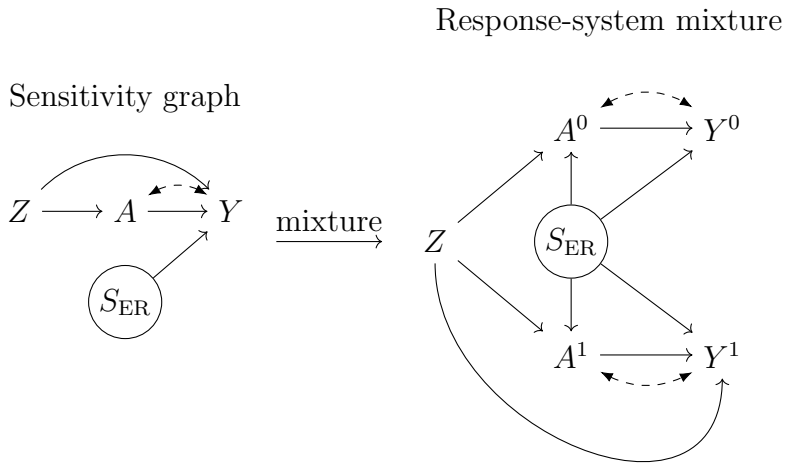
Let I_λ *be the indicator of the strata that violate an individual-level restriction. Suppose the selected response system cannot violate the restriction when* $S_\lambda = 0$*. For a population-level restriction,* S_λ *is an augmented component indicator and the restriction is imposed on the conditional response distribution given* $S_\lambda = 0$*, when that component has positive probability mass. The population distribution must satisfy all retained restrictions and observed data.*

For the particular case of individual-level restrictions, I_λ indicates the generalized principal strata that violate the restriction, so $P(I_\lambda = 1)$ is their exact population proportion. The sensitivity indicator S_λ identifies the unrestricted component, whose proportion is $P(S_\lambda = 1)$. Since all violating strata must belong to this component, but nonviolating strata may also belong to it, $P(I_\lambda = 1) \leq P(S_\lambda = 1) \leq \theta_\lambda^{\max}$.

Figure 4 illustrates a sensitivity graph for an instrumental variable model with a relaxed exclusion restriction. The violation indicator is determined by the outcome strata. Because treatment and outcome belong to the same district, their strata may be dependent, so the indicator may also be dependent on the treatment strata. The exogeneity assumption makes Z independent of the joint treatment and outcome strata. Conditioning the response law on this violation, then, preserves instrument independence. The doubled representation, defined below, draws the two branch systems independently of the selector and evaluates the selected system using an independent instrument with the observed distribution. The selector chooses the entire $\{A, Y\}$ response system, although the restriction is located at Y in the sensitivity graph. The restricted branch, because it imposes exclusion restriction,

omits the arrow $Z \to Y$. The unrestricted branch, on the other hand, allows it. The selector weight and the branch support restrictions cannot be derived from the ordinary edges alone and must be specified separately.

Figure 4: **Sensitivity graph and mixture representation for IV exclusion.** The left panel locates the restriction at Y. The right panel shows the doubled representation with restricted and unrestricted response parts sharing the independent instrument input Z. Directed arrows from S_{ER} indicate selection of the relaxed or original response system. This representation preserves all the retained constraints from the original model.



B.3 The enlarged doubled representation

The doubled representation makes the selection rule explicit. Let $\mathbf{R}^0$ and $\mathbf{R}^1$ be complete generalized principal strata in the restricted and unrestricted branches. The equivalence of the program derived by canonicalization of this ADMG with the sensitivity problem that relaxes an individual-level restriction is proved in Appendix B.3. This is a mixture construction. Appendix B.5 extends this construction to many selectors.

The doubled representation is sound in the sense that every recovered model satisfies the original sensitivity constraints and the other stated restrictions, including $P(I_\lambda = 1) \leq P(S_\lambda = 1) \leq \theta_\lambda^{\max}$.

Proposition 1 (Soundness of the doubled representation for individual-level restrictions)**.** *Let* $\mathbf{R}^0$ *and* $\mathbf{R}^1$ *be the complete generalized principal strata in the restricted and relaxed copies, and let* I_λ *indicate a violation of the functional restriction. Suppose*

$$I_\lambda(\mathbf{R}^0) = 0 \quad \textit{almost surely}, \qquad P(S=1) \leq \theta_\lambda^{\max}.$$

Recover the original strata by $\mathbf{R} = \mathbf{R}^S$*, using the same* S *and impose the other original assumptions and observed data constraints on the recovered model. Then the set of recovered models is a subset of the original sensitivity model class.*

Proof. The restricted copy contains no violations, so

$$\begin{aligned} P\{I_\lambda(\mathbf{R}) = 1\} &= P\{S=0, I_\lambda(\mathbf{R}^0) = 1\} + P\{S=1, I_\lambda(\mathbf{R}^1) = 1\} \\ &= P\{S=1, I_\lambda(\mathbf{R}^1) = 1\} \\ &\leq P(S=1) \leq \theta_\lambda^{\max}. \end{aligned}$$

As all the retained assumptions are being imposed, this is sufficient to prove soundness. □

It remains to be proved that the doubled representation is also complete, i.e. if the original sensitivity model is a subset of the recovered models. As that is the case, the doubled representation is equivalent to the original sensitivity model for individual-level restrictions.

Proposition 2 (Equivalence of the enlarged doubled program)**.** *Let* r *denote a generalized principal stratum, and let* $I_\lambda(r)$ *equal 1 if that stratum violates the restriction and 0 otherwise. Let* $p_r(\boldsymbol{\eta})$ *be the probability of the strata under the original model. Assume that the set* $\{r : I_\lambda(r) = 0\}$ *is nonempty. Keep* $\boldsymbol{\eta} \in \mathcal{H}_0(P_V)$*, which imposes the observed data constraints and all original assumptions except the restriction being relaxed. Replace the original violation*

bound $P_{\boldsymbol{\eta}}(\mathcal{V}_\lambda) \le \theta_\lambda^{\max}$ *by the following constraints:*

$$
\begin{aligned}
p_r(\boldsymbol{\eta}) &= (1-t)q_r^0 + tq_r^1 && \text{for every } r,\\
q_r^s \ge 0, \qquad \sum_r q_r^s &= 1 && s = 0, 1,\\
q_r^0 &= 0 && \text{if } I_\lambda(r) = 1,\\
0 \le t &\le \theta_\lambda^{\max}.
\end{aligned}
$$

The original independence restrictions are imposed on the combined distribution, not separately on q^0 *and* q^1*. The values of* $\boldsymbol{\eta}$ *compatible with these constraints are exactly*

$$\{\boldsymbol{\eta} \in \mathcal{H}_0(P_V) : \sum_r I_\lambda(r) p_r(\boldsymbol{\eta}) \le \theta_\lambda^{\max}\}.$$

Therefore, solving the global optimization problem with the original estimand $\mathcal{E}(\boldsymbol{\eta})$ *as the objective function gives the same bounds in the two programs.*

Proof. Soundness follows from Proposition 1. For equivalence, then, it remains to prove completeness.

Take any feasible original $\boldsymbol{\eta}$ and set $t = \sum_r I_\lambda(r) p_r(\boldsymbol{\eta})$. If $0 < t < 1$, choose q^0 and q^1 to be the distributions over strata conditional on $I_\lambda = 0$ and $I_\lambda = 1$, respectively. This implies

$$
\begin{aligned}
q_r^s \ge 0, \qquad \sum_r q_r^s &= 1 && s = 0, 1,\\
q_r^0 &= 0 && \text{if } I_\lambda(r) = 1.
\end{aligned}
$$

For $0 < t < 1$, the law of total probability guarantees

$$p_r(\boldsymbol{\eta}) = (1-t)q_r^0 + tq_r^1 \qquad \text{for every } r.$$

For $t = 0$, conditioning on $I_\lambda = 1$ is undefined, so set $q_r^0 = p_r(\boldsymbol{\eta})$ and choose any probability

distribution for q^1. Analogously, for $t = 1$, conditioning on $I_\lambda = 0$ is undefined, so set $q_r^1 = p_r(\boldsymbol{\eta})$ and choose any probability distribution for q^0 that assigns probability 0 to all violating strata.

Finally, draw the strata in copies 0 and 1 independently with probabilities q_r^0 and q_r^1, and independently draw S with $P(S = 1) = t$. Select the stratum from copy S, keeping this choice fixed across interventions. The probability of selecting stratum r is $(1 - t)q_r^0 + tq_r^1 = p_r(\boldsymbol{\eta})$, so this construction reproduces the original strata probabilities. That proves completeness and equivalence. □

B.4 Augmented graphs

Definition 10 (Augmented model)**.** *Let $\mathcal{M}_R$ denote a relaxed causal model, and let C_λ be a population-level constraint for a natural sensitivity analysis problem. An* augmented model *M_A is a causal model constructed by appending a binary auxiliary variable H to the original model $\mathcal{M}_R$, where the distribution conditional on $H = 0$ satisfies the population-level constraint C_λ, whereas no such requirement is imposed on the distribution conditional on $H = 1$.*

Lemma 3 (Invariance under unrestricted augmentation)**.** *Fix one district of a finite causal model and form an augmented enlargement with indicator H. If no observed data restriction, structural assumption, or estimand involves H, then projecting the augmented model onto the original variables gives exactly the original feasible set of distributions. Therefore, the identified set, and the respective sharp bounds, for any estimand defined on the original variables remain invariant.*

Proof of Lemma 3. Firstly, take any data probability distribution in the augmented model and marginalize over H. As H has no children, the structural equations and principal strata of the original variables are the same. At last, as there is no constraint involving H, the resulting marginal distribution belongs to the original model.

Any feasible distribution in the original model can be extended by setting $H = 0$, without changing the original variables. As the estimand does not involve H, the two models have the same feasible estimand values and sharp bounds. □

Lemma 1 (Individual-level representation under augmentation)**.** *Assume that C_λ holds within the $H = 0$ component of the augmented model. Then imposing $P(H = 1) = 0$ recovers exactly the original model that satisfies C_λ, by projection onto the original variables. Therefore, relaxing this individual-level restriction to $P(H = 1) \leq \theta_\lambda^{\max}$ defines a sensitivity problem to which Theorem 1 and its corollaries apply under their stated conditions.*

Proof of Lemma 1. Assume $P(H = 1) = 0$. Then the total population belongs to the $H = 0$ component, so its probability distribution satisfies C_λ. Projection does not change the original data constraints or the retained model restrictions. Any probability distribution in the original model satisfying C_λ can also be augmented by setting $H = 0$ almost surely. This does not change those constraints and satisfies C_λ within $H = 0$, proving equality of the projected feasible sets.

The restriction $P(H = 1) = 0$ excludes all augmented strata with $H = 1$ and is therefore individual-level. Keeping the requirement that C_λ hold within $H = 0$ among the constraints in $\mathcal{H}_0(P_V)$, apply Theorem 1 with violation event $\{H = 1\}$. Finally, the corollaries of the theorem also apply under their stated conditions. □

Lemma 4 (From an augmented indicator to an independent selector)**.** *Let R denote the complete generalized principal stratum and let H be a binary augmented indicator, with any joint distribution $P(H, R)$. There exist two branch strata R^0, R^1 and a binary selector S, independent of (R^0, R^1), such that:*

$$P(S = s, R^S = r) = P(H = s, R = r) \qquad \textit{for every } s, r.$$

So, the doubled representation does not change the dependence between the indicator and the selected strata, all constraints on their joint probabilities, and any population-level restriction

conditional on $H = 0$, which becomes conditional on $S = 0$ when this event has positive probability.

Proof. Let $P(S = s) = P(H = s)$. If $P(H = s) = 0$, choose any distribution over strata for R^s. For each s with $P(H = s) > 0$, let R^s have probabilities $P(R^s = r) = P(R = r \mid H = s)$. Draw the strata for each branch independently of each other and of S, and select R^S.

Now, if $P(H = s) = 0$,$P(S = s, R^S = r) = P(H = s, R = r) = 0$. And for each s with $P(H = s) > 0$,

$$\begin{aligned} P(S = s, R^S = r) &= P(S = s, R^s = r) \\ &= P(S = s)P(R^s = r) \\ &= P(H = s)P(R = r \mid H = s) \\ &= P(H = s, R = r). \end{aligned}$$

Therefore, $P(S = s, R^S = r) = P(H = s, R = r)$ for every s, r. Then, replacing (H, R) by (S, R^S) does not change any constraint on these probabilities or any estimand calculated from them. Any doubled distribution also defines an augmented distribution by setting $P(H = s, R = r) = P(S = s, R^S = r)$. And the two representations give the same feasible distributions and estimand values, proving the statement. □

Figure 5 illustrates the augmented construction for unconfoundedness and population-level monotonicity. In both cases, the selector is independent of the two branches, but may be dependent on the selected strata.

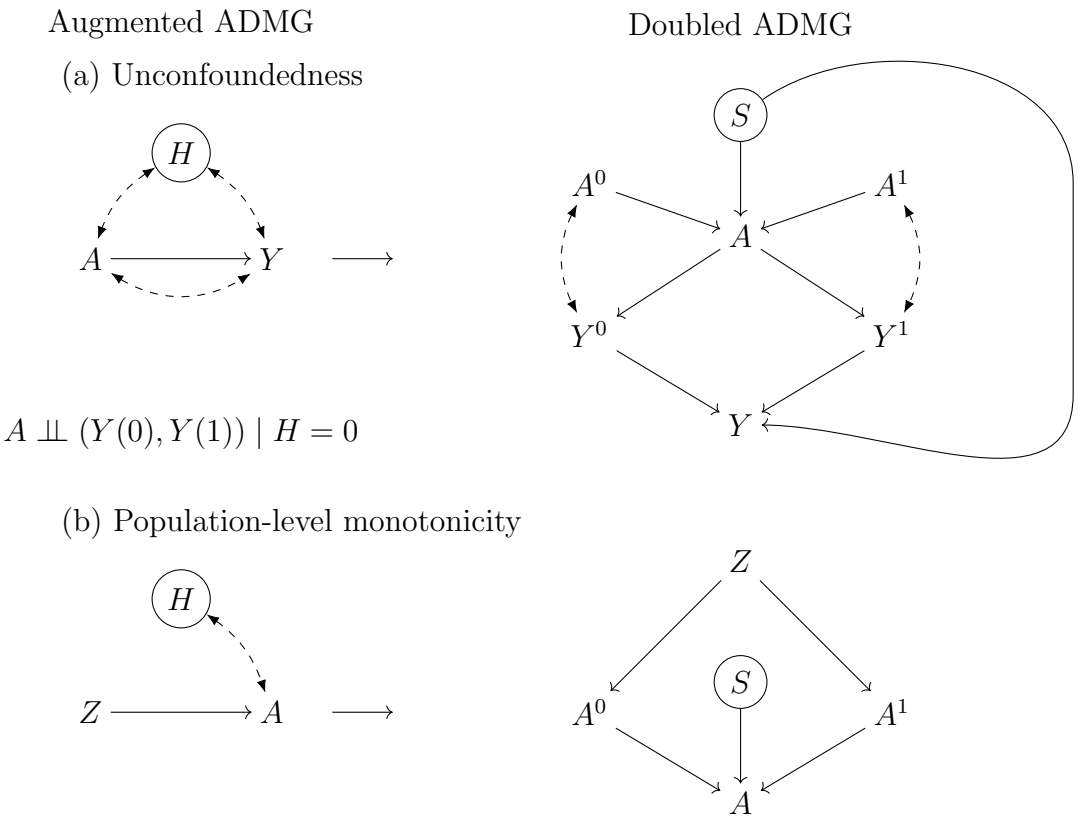


Figure 5: **Augmented and doubled ADMGs for population-level restrictions.** Circles denote the latent component indicators H and S. Bidirected edges represent dependence through unobserved common causes. On the left, H may depend on the original strata. On the right, branch s has the original strata distribution conditional on $H = s$. The branch strata and S are drawn independently, with $P(S = 1) = P(H = 1) \leq \theta^{\max}$. In (a), $A = A^S$, each Y^s is evaluated at this selected treatment, and $Y = Y^S$. So, branch 0 satisfies unconfoundedness as a restriction on its strata probabilities. In (b), A^s is evaluated at the common independent instrument Z, $A = A^S$, and branch 0 satisfies $\mathbb{E}[A^0(z_1) - A^0(z_0)] \geq 0$. These branch restrictions and the deterministic selection equations are imposed in addition to the ADMGs. All retained constraints apply to the selected population distribution.

Lemma 5 (Unconfoundedness as a population-level constraint)**.** *For finite-valued A and Y, unconfoundedness $C_{\mathrm{UC}} : A \perp\!\!\!\perp \mathbf{Y}_A$ is a population-level constraint in Definition 6. In the augmented representation, when $q_{0++} > 0$, $A \perp\!\!\!\perp \mathbf{Y}_A \mid H = 0$ is equivalent to*

$$q_{0i\boldsymbol{r}} q_{0++} = q_{0i+} q_{0+\boldsymbol{r}} \quad \text{for every } a_i, \mathbf{y}_{\boldsymbol{r}}, \tag{3}$$

where a $+$ denotes summation over the corresponding coordinate. When $q_{0++} = 0$, these equations are vacuous. Its component relaxation is therefore represented by these polynomial equalities together with $\sum_{i,\boldsymbol{r}} q_{1i\boldsymbol{r}} = P(H = 1) \le \theta_{\mathrm{UC}}^{\max}$, with no unconfoundedness restriction on the $H = 1$ component.

Proof of Lemma 5. Write $p_{i\boldsymbol{r}} = P(a_i \mathbf{y}_{\boldsymbol{r}})$, with a $+$ denoting summation over the corresponding coordinate. Unconfoundedness is equivalent to $p_{i\boldsymbol{r}} = p_{i+} p_{+\boldsymbol{r}}$ for every $i, \boldsymbol{r}$, or to the single polynomial constraint

$$h_{\mathrm{UC}}(\mathbf{p}) = \sum_{i,\boldsymbol{r}} (p_{i\boldsymbol{r}} - p_{i+} p_{+\boldsymbol{r}})^2 = 0.$$

Thus it is population-level under Definition 6.

If $q_{0++} > 0$, divide Equation (3) by q_{0++}^2 to obtain

$$P(a_i \mathbf{y}_{\boldsymbol{r}} \mid H = 0) = P(a_i \mid H = 0) P(\mathbf{y}_{\boldsymbol{r}} \mid H = 0).$$

Thus the polynomial equalities imply $A \perp\!\!\!\perp \mathbf{Y}_A \mid H = 0$. Multiplying this conditional factorization by q_{0++}^2 proves the converse part. If $q_{0++} = 0$, the equalities are vacuous and the conditional distribution on the null component can be arbitrarily chosen.

Marginalizing H gives $P(a_i \mathbf{y}_{\boldsymbol{r}}) = q_{0i\boldsymbol{r}} + q_{1i\boldsymbol{r}}$. Finally, summing the $H = 1$ cells gives $P(H = 1) = \sum_{i,\boldsymbol{r}} q_{1i\boldsymbol{r}}$, which proves the sensitivity statement.

The mixture representation follows directly. Write $t = P(H = 1)$. For $0 < t < 1$, define

the probability distributions

$$\alpha_i = \frac{q_{0i+}}{1-t}, \qquad \beta_{\boldsymbol{r}} = \frac{q_{0+\boldsymbol{r}}}{1-t}, \qquad \rho_{i\boldsymbol{r}} = \frac{q_{1i\boldsymbol{r}}}{t}.$$

The factorization gives $q_{0i\boldsymbol{r}} = (1-t)\alpha_i\beta_{\boldsymbol{r}}$, so marginalizing H yields

$$P(a_i\mathbf{y}_{\boldsymbol{r}}) = (1-t)\alpha_i\beta_{\boldsymbol{r}} + t\rho_{i\boldsymbol{r}}.$$

On the other hand, any such mixture defines augmented cells $q_{0i\boldsymbol{r}} = (1-t)\alpha_i\beta_{\boldsymbol{r}}$ and $q_{1i\boldsymbol{r}} = t\rho_{i\boldsymbol{r}}$, which satisfy Equation (3). At $t = 0$ or $t = 1$, choose the unused component distributions arbitrarily. Therefore imposing $t \leq \theta_{\mathrm{UC}}^{\max}$ gives precisely the contamination mixture with the stated weight bound.

□

B.5 Multiple sensitivity nodes and overlapping relaxations

It is possible to relax many different assumptions by extending the doubled representation with many binary selectors $S_1, \ldots, S_J$. In particular, the model implies up to 2^J branches, with branch s retaining restriction j when $s_j = 0$ and allowing its violation when $s_j = 1$. Let

$$S = (S_1, \ldots, S_J), \qquad \pi_s = P(S = s), \qquad s \in \{0,1\}^J.$$

The joint distribution allows for the possibility of overlap among the relaxations, which typically happens when strata are constrained by multiple selectors.

I first prove that multiple relaxations of individual-level assumptions have an equivalent mixture representation. I then show that, under the stated assumptions on sensitivity graphs, recursively doubling the graphs produces an ADMG with up to 2^J copies. Finally, I extend these results to relaxations of population-level assumptions.

Proposition 3 (Mixture representation of multiple relaxations)**.** *Let $I_1(r), \ldots, I_J(r)$ be individual-level violation indicators on the complete states of a common finite canonical model. Retain all observed-data and other model constraints on the recovered distribution $p_r(\boldsymbol{\eta})$. For each $s \in \{0,1\}^J$, introduce a component distribution q^s supported on states satisfying $I_j(r) = 0$ whenever $s_j = 0$. Empty branches receive zero weight and are omitted. Then the program*

$$
\begin{aligned}
p_r(\boldsymbol{\eta}) &= \sum_{s \in \{0,1\}^J} \pi_s q_r^s, && \text{for every } r, \\
\pi_s \geq 0, \qquad & \sum_s \pi_s = 1, \\
\sum_{s: s_j = 1} \pi_s &\leq \theta_j^{\max}, && j = 1, \ldots, J,
\end{aligned}
$$

with each nonempty component normalized, has exactly the same feasible recovered distributions as the simultaneous sensitivity model of Corollary 3.

Proof. For any feasible mixture, the support restrictions imply

$$
\sum_r I_j(r) p_r(\boldsymbol{\eta}) = \sum_s \pi_s \sum_r I_j(r) q_r^s \leq \sum_{s: s_j = 1} \pi_s \leq \theta_j^{\max}.
$$

The remaining model constraints hold by construction. This suffices to prove that the feasible mixture program produces a distribution satisfying the sensitivity bounds.

To prove the converse implication, take a feasible original distribution and let π_s be the probability that its vector of violation indicators equals s. For each branch with positive probability, choose q^s as the conditional probability distribution given that vector. Then the mixture equations follow from the law of total probability, and the total probability of branches with $s_j = 1$ is equal to the corresponding violation proportion. A nonempty branch with zero probability can use any distribution on its allowed support. □

Proposition 4 (Recursive graphical representation)**.** *For each selector configuration $s \in$*

$\{0,1\}^J$, let $\mathcal{G}_s$ be a corresponding branch ADMG. Suppose the directed edges of all the branch ADMGs have an acyclic union. Assume that each component distribution q^s admits a causal representation that is compatible with its branch ADMG $\mathcal{G}_s$.

An expanded ADMG can be constructed from disjoint copies of the branch ADMGs, with one variable copy V^s for each branch and selector arrows entering the affected variables. Proposition 2 establishes the corresponding equivalence with a mixture of two components. Dependence among selectors might be represented by a common latent parent.

From the expanded ADMG, one can construct a sensitivity graph by merging variable copies and keeping all branch edges and selector arrows. The component probability distributions must allow any additional dependence induced by conditioning on selectors. All the other constraints on the recovered distribution are kept.

Proof. To construct the expanded ADMG, first for each branch s, take a separate copy of its causal representation over a variable set V^s. Let the branch states be independent across copies, and add the elements of the selector vector with joint probability π, independently of all branch states. The original variables can be recovered by selecting the same branch s for all potential outcomes of a particular unit. So the recovered probability distribution is $\sum_s \pi_s q^s$.

Now construct the expanded graph from the disjoint ADMGs and add the selector arrows into the affected variables in each copy. It remains to show that this graph is acyclic. First, each branch copy has an acyclic directed part, and there are no directed edges between the distinct copies. Second, arrows from selectors point to the copies, but no directed arrow from a copy points to a selector. At last, there are no directed edges among selectors. Dependence among selectors can be represented by a common latent parent whose only children are the selectors. Its latent projection introduces bidirected edges among selectors without changing the directed part of the graph. Therefore, the expanded graph is acyclic.

Now to construct the sensitivity graph, merge corresponding variable copies in the expanded ADMG, keeping all branch edges and selector arrows. Then, the resulting graph will

have no directed cycles because the union of branch directed edges is acyclic and no directed arrow from a branch copy points to a selector.

Finally, adding the retained constraints on the recovered probability distribution guarantees that the recovered distribution satisfies the observed data and other model requirements.

□

Corollary 4 (Mixture representation of population-level relaxations)**.** *Suppose the population-level constraints are jointly represented in a common finite augmented canonical model, as in Lemma 1. Then, Proposition 3 gives an equivalent mixture representation of their simultaneous relaxations. All retained constraints, including the population-level ones in the augmented components, are imposed on the recovered augmented distribution, rather than separately in every selector branch.*

Proof. Let H_j be the augmented indicator for restriction j. On each complete augmented state r, define $I_j(r) = \mathbf{1}\{H_j(r) = 1\}$. The relaxation bounds are then bounds on individual-level violation indicators. Applying Proposition 3 to these indicators and retaining all other constraints on the recovered augmented distribution suffices to prove the result. □

Different assumptions may be relaxed for the same units. The joint probability distribution of the selectors allows this overlap without requiring the selectors to be independent or mutually exclusive. Consider two relaxations, and let $\theta_j = P(S_j = 1)$ and $\omega = P(S_1 = 1, S_2 = 1)$. Their overlap satisfies the Frechet bounds:

$$\max\{0, \theta_1 + \theta_2 - 1\} \leq \omega \leq \min\{\theta_1, \theta_2\}.$$

Any value in those inequalities is feasible for the overlap of the two relaxations. Assume first the selectors are mutually independent, then their overlap is $\omega = \theta_1\theta_2$. In fact, imposing it can restrict the sensitivity model. Mutual exclusivity, on the other hand, is different. It requires $\pi_s = 0$ whenever more than one coordinate of s is one, a support restriction that an ordinary ADMG alone does not encode.

Furthermore, the mixture representation is invariant to the order in which restrictions are listed, if the branch definitions and all compatibility constraints are fixed on the same expanded model. Reordering merely permutes the coordinates of s and the corresponding proportions and probability distributions but does not change $\sum_s \pi_s q_r^s$. For example, jointly relaxing no defiers and exclusion in the IV model gives four branches: a) both restrictions retained, b) only no defiers relaxed, c) only exclusion relaxed, d) and both relaxed.

Alternatively, a single selector can distinguish a component satisfying all restrictions from one in which all are relaxed. This gives a one-dimensional analysis, with the proportion of the latter component bounded by a single sensitivity parameter. For individual-level restrictions, the corresponding compound violation event is their union. In the instrumental variable case, that would mean for the mixture representation a branch between a model satisfying at the same time exclusion restriction and monotonicity and another branch, relaxing both. For two restrictions, the probability is

$$P(I_1 = 1 \text{ or } I_2 = 1) = P(I_1 = 1) + P(I_2 = 1) - P(I_1 = 1, I_2 = 1).$$

Therefore, this analysis does not in general coincide with assigning the same marginal bound to each separate selector.

C Results for Standard Sensitivity Scenarios

C.1 Symbolic sensitivity function for simple confounding

Theorem 5 (Natural sensitivity bounds under classic confounding)**.** *Let $\theta = \theta_{\mathrm{UC}} \in [0,1]$ be the nominal weight of the component in which confounding is permitted, and suppose the mixture is compatible with the observed distribution. Write $p_{ay} = P(A = a, Y = y)$ and $t_{ha} = P(H = h, A = a)$, as in Appendix C.1. The feasible range of t_{0a} is*

$$t_{0a}^{-} = \max\{0, P(A = a) - \theta\} \leq t_{0a} \leq \min\{1 - \theta, P(A = a)\} = t_{0a}^{+}.$$

At each fixed $t_{0a} > 0$, define

$$L_{P(Y(A=a)=y)}(t_{0a}) = p_{ay} + \max\left\{0, \frac{(1-\theta-t_{0a})(t_{0a}-p_{a,1-y})}{t_{0a}}\right\},$$
$$U_{P(Y(A=a)=y)}(t_{0a}) = \theta + t_{0a} - p_{a,1-y} + \min\left\{1-\theta-t_{0a}, \frac{(1-\theta-t_{0a})p_{ay}}{t_{0a}}\right\}.$$

At $t_{0a} = 0$, these bounds are p_{ay} and $1 - p_{a,1-y}$, respectively. The sharp marginal bounds are the minimum of L and maximum of U over the two endpoints t_{0a}^-, t_{0a}^+.

For the ATE, let $L_a(x) = L_{P(Y(A=a)=1)}(x)$ and $U_a(x) = U_{P(Y(A=a)=1)}(x)$. Both potential outcomes must use the same allocation $t = t_{01}$, with $t_{00} = 1 - \theta - t$. Define

$$L_{\mathrm{ATE}}(t) = L_1(t) - U_0(1-\theta-t), \qquad U_{\mathrm{ATE}}(t) = U_1(t) - L_0(1-\theta-t).$$

Let $\mathcal{T} = [t_{01}^-, t_{01}^+]$ and

$$\mathcal{B}_L = \{t_{01}^-, t_{01}^+, p_{10}, 1-\theta-p_{01}\} \cap \mathcal{T}, \qquad \mathcal{B}_U = \{t_{01}^-, t_{01}^+, p_{11}, 1-\theta-p_{00}\} \cap \mathcal{T}.$$

The sharp ATE sensitivity bounds are

$$\mathcal{L}_{\mathrm{ATE}}(\theta) = \min_{t\in\mathcal{B}_L} L_{\mathrm{ATE}}(t), \qquad \mathcal{U}_{\mathrm{ATE}}(\theta) = \max_{t\in\mathcal{B}_U} U_{\mathrm{ATE}}(t).$$

The boundary conventions above apply when $t = 0$ or $1 - \theta - t = 0$.

Proof. The bounds over the full feasible interval and their sharpness follow from Theorems 6, 7, and 8. Theorem 7 reduces the marginal optimization to the feasible endpoints, with the other treatment and outcome values obtained by relabeling. Theorem 8 reduces the ATE optimization to the candidate sets $\mathcal{B}_L$ and $\mathcal{B}_U$. □

If $P(A = a) > 0$, $\theta = 0$ gives the point-identified value $P(Y = y \mid A = a)$, while $\theta = 1$ gives the usual no assumption bounds $[p_{ay}, 1 - p_{a,1-y}]$. If $P(A = a) = 0$ and $\theta = 0$, the target instead remains unrestricted in $[0, 1]$.

C.1.1 Setting the optimization problem

For binary treatment A and outcome Y, write $h_h a_i y_{rs}$ for the augmented stratum satisfying $H = h, A = i, (Y(0), Y(1)) = (r, s)$, where all indices are binary. The h_0 strata form the unconfounded component, and the h_1 strata form the unrestricted component. Write

$\theta = \theta_{\text{UC}} = P(H = 1)$. By consistency, the observed cells are

$$
\begin{aligned}
P(A = 1, Y = 1) &= P(h_0 a_1 y_{01}) + P(h_0 a_1 y_{11}) + P(h_1 a_1 y_{01}) + P(h_1 a_1 y_{11}),\\
P(A = 1, Y = 0) &= P(h_0 a_1 y_{00}) + P(h_0 a_1 y_{10}) + P(h_1 a_1 y_{00}) + P(h_1 a_1 y_{10}),\\
P(A = 0, Y = 1) &= P(h_0 a_0 y_{10}) + P(h_0 a_0 y_{11}) + P(h_1 a_0 y_{10}) + P(h_1 a_0 y_{11}),\\
P(A = 0, Y = 0) &= P(h_0 a_0 y_{00}) + P(h_0 a_0 y_{01}) + P(h_1 a_0 y_{00}) + P(h_1 a_0 y_{01})
\end{aligned}
$$

The stratum probabilities are nonnegative and sum to one, with $\sum_{i,r,s} P(h_1 a_i y_{rs}) = \theta$. Unconfoundedness within $H = 0$ imposes

$$
(1 - \theta_{\text{UC}}) P(h_0 a_i y_{rs}) = \left\{ \sum_{r',s'} P(h_0 a_i y_{r's'}) \right\} \left\{ \sum_{i'} P(h_0 a_{i'} y_{rs}) \right\} \quad \text{for every } i, r, s
$$

The first objective function is $P(Y(A = 1) = 1)$. The translation of this estimand into strata shows that it is composed of a point identified part, $P(A = 1, Y = 1)$, and a partially identified part, $P(H = 0, A = 0, Y(A = 1) = 1)$ and $P(H = 1, A = 0, Y(A = 1) = 1)$, so bounds on the latter are sufficient. In particular, the partially identified part has quantities that belong respectively to the unconfounded and the confounded arms.

$$
\begin{aligned}
P(Y(A = 1) = 1) &= \sum_{k=0}^{1} \sum_{i=0}^{1} \left[P(h_k a_i y_{01}) + P(h_k a_i y_{11}) \right] =\\
&\underbrace{P(h_0 a_1 y_{01}) + P(h_0 a_1 y_{11}) + P(h_1 a_1 y_{01}) + P(h_1 a_1 y_{11})}_{P(A=1,Y=1)}\\
&+ \underbrace{P(h_0 a_0 y_{01}) + P(h_0 a_0 y_{11})}_{P(H=0,A=0,Y(A=1)=1)} + \underbrace{P(h_1 a_0 y_{01}) + P(h_1 a_0 y_{11})}_{P(H=1,A=0,Y(A=1)=1)}.
\end{aligned}
$$

We bound both contributions at a fixed treatment allocation and then optimize their sum over all feasible allocations.

For the ATE, the strata with $Y(0) = Y(1)$ cancel, leaving

$$\begin{aligned}
\text{ATE} &= P(Y(1) = 1) - P(Y(0) = 1) \\
&= \sum_{k=0}^{1} \sum_{i=0}^{1} \left[P(h_k a_i y_{01}) - P(h_k a_i y_{10}) \right] \\
&= P(h_0 a_0 y_{01}) + P(h_0 a_1 y_{01}) - P(h_0 a_0 y_{10}) - P(h_0 a_1 y_{10}) \\
&\quad + P(h_1 a_0 y_{01}) + P(h_1 a_1 y_{01}) - P(h_1 a_0 y_{10}) - P(h_1 a_1 y_{10}).
\end{aligned}$$

There is still an identified contribution from the observed outcomes. By consistency, the same objective can be written as

$$\begin{aligned}
\text{ATE} = \underbrace{P(A = 1, Y = 1) - P(A = 0, Y = 1)}_{\text{identified contribution}} \\
+ \sum_{k=0}^{1} \left[P(h_k a_0 y_{01}) + P(h_k a_0 y_{11}) \right] \\
- \sum_{k=0}^{1} \left[P(h_k a_1 y_{10}) + P(h_k a_1 y_{11}) \right].
\end{aligned}$$

The global solutions for these optimization problems are at the same time the perturbation and the sensitivity functions, because the former is suitably monotonic, as shown in Theorem 9.

C.1.2 Bounds for $P(Y(A = 1) = 1)$ and the ATE

The two nonidentified elements can be optimized. Call $p_u = P(H = 0, A = 0, Y(A = 1) = 1)$ and $p_c = P(H = 1, A = 0, Y(A = 1) = 1)$.

The idea is to characterize the feasible set in parts. Denote $t_{ha} = P(H = h, A = a)$. Then let $t_{01} = P(H = 0, A = 1) = \sum_{r,s} P(h_0 a_1 y_{r,s})$. The feasible values for t are upper-bounded by $P(H = 0) = 1 - \theta$ and $P(A = 1)$ by containment. Analogously, $P(H = 0, A = 1) =$

$P(A=1) - P(H=1, A=1) \geq P(A=1) - \theta$, so:

$$\max\{0, P(A=1) - \theta\} \leq t_{01} \leq \min\{1-\theta, P(A=1)\}$$

Notice that by fixing t_{01}, then the other t-quantities are immediately identifiable:

$$t_{11} = P(A=1) - t_{01}, \quad t_{00} = 1 - \theta - t_{01}, \quad t_{10} = \theta + t_{01} - P(A=1)$$

Before targeting p_u, consider $P(H=0, A=1, Y(A=1)=1)$, which is equal to $P(H=0, A=1, Y=1)$, by consistency. Denote $P(H=h, A=a, Y=y)$ by q_{hay}, so $q_{011} = P(H=0, A=1, Y(A=1)=1)$. By containment, q_{011} is upper-bounded by t_{01} and $P(A=1, Y=1)$. Analogously, $q_{011} = P(A=1, Y=1) - q_{111} \geq P(A=1, Y=1) - \min\{P(A=1, Y=1), t_{11}\}$. As t_{11} is a function of t_{01}:

$$\max\{0, t_{01} - P(A=1, Y=0)\} \leq q_{011} \leq \min\{t_{01}, P(A=1, Y=1)\} \tag{4}$$

With fixed $q_{011}, t_{00}, t_{01}, t_{10}, t_{11}$, it is possible to consider the bounds for p_u. Unconfoundedness implies the identity

$$p_u t_{01} = q_{011} t_{00}.$$

When both treatment-group masses are positive, this is equivalent to $P(Y(A=1)=1 \mid H=0, A=1) = P(Y(A=1)=1 \mid H=0, A=0)$. Assume first t_{01} is positive. The cases where $t_{01}=0$ will be handled separately.

For further simplification, denote $p_{ay} = P(A=a, Y=y)$. Substituting into Equation 4:

$$\frac{t_{00}}{t_{01}} \cdot \max\{0, t_{01} - p_{10}\} \leq p_u \leq \frac{t_{00}}{t_{01}} \cdot \min\{t_{01}, p_{11}\}$$

Substituting $t_{00} = 1 - \theta - t_{01}$ and rearranging gives:

$$\max\left\{0, \frac{(1-\theta-t_{01})(t_{01}-p_{10})}{t_{01}}\right\} \leq p_u \leq \min\left\{(1-\theta-t_{01}), \frac{(1-\theta-t_{01})p_{11}}{t_{01}}\right\} \quad (5)$$

Now consider p_c. By containment, $0 \leq p_c \leq t_{10}$. As t_{10} is a function of t_{01}:

$$0 \leq p_c \leq \theta + t_{01} - P(A=1) \quad (6)$$

That is sufficient to show that p_u and p_c can be fully bounded considering the feasible values of t_{01}. Then the bounds for $P(Y(A=1)=1)$ can be computed by summing the bounds of p_u, p_c, and the identified part, which is equal to p_{11}.

For a particular t_{01}, the bounds $L_{P(Y(A=1)=1)}(t_{01})$ and $U_{P(Y(A=1)=1)}(t_{01})$ are:

$$L_{P(Y(A=1)=1)}(t_{01}) = p_{11} + \max\left\{0, \frac{(1-\theta-t_{01})(t_{01}-p_{10})}{t_{01}}\right\}$$
$$U_{P(Y(A=1)=1)}(t_{01}) = \theta + t_{01} - p_{10} + \min\left\{1-\theta-t_{01}, \frac{(1-\theta-t_{01})p_{11}}{t_{01}}\right\}$$

For the case $t_{01} = 0$, then $L_{P(Y(A=1)=1)}(0) = p_{11}$ and $U_{P(Y(A=1)=1)}(0) = 1 - p_{10}$.

Notice that by changing the value of A from 1 to 0, the bounds for $P(Y(A=0)=1)$, when $t_{00} > 0$, are:

$$L_{P(Y(A=0)=1)}(t_{00}) = p_{01} + \max\left\{0, \frac{(1-\theta-t_{00})(t_{00}-p_{00})}{t_{00}}\right\}$$
$$U_{P(Y(A=0)=1)}(t_{00}) = \theta + t_{00} - p_{00} + \min\left\{1-\theta-t_{00}, \frac{(1-\theta-t_{00})p_{01}}{t_{00}}\right\}$$

At $t_{00} = 0$, the corresponding bounds are $L_{P(Y(A=0)=1)}(0) = p_{01}$ and $U_{P(Y(A=0)=1)}(0) = 1 - p_{00}$.

To calculate the bounds for the ATE, first reconstruct the bounds for $P(Y(A=0)=1)$

as a function of t_{01}, using the fact that $t_{00} = 1 - \theta - t_{01}$:

$$L_{P(Y(A=0)=1)}(t_{01}) = p_{01} + \max\left\{0, \frac{(t_{01})(1-\theta-t_{01}-p_{00})}{1-\theta-t_{01}}\right\}$$
$$U_{P(Y(A=0)=1)}(t_{01}) = 1 - t_{01} - p_{00} + \min\left\{t_{01}, \frac{(t_{01})p_{01}}{1-\theta-t_{01}}\right\}$$

At last, for a particular t_{01}, valid bounds for the ATE can be calculated as:

$$\begin{aligned} L_{\text{ATE}}(t_{01}) &= L_{P(Y(A=1)=1)}(t_{01}) - U_{P(Y(A=0)=1)}(t_{01}) \\ &= p_{11} + p_{00} + t_{01} - 1 + \max\left\{0, \frac{(1-\theta-t_{01})(t_{01}-p_{10})}{t_{01}}\right\} \\ &\quad - \min\left\{t_{01}, \frac{t_{01}p_{01}}{1-\theta-t_{01}}\right\}, \end{aligned}$$

$$\begin{aligned} U_{\text{ATE}}(t_{01}) &= U_{P(Y(A=1)=1)}(t_{01}) - L_{P(Y(A=0)=1)}(t_{01}) \\ &= \theta + t_{01} - p_{10} - p_{01} + \min\left\{1-\theta-t_{01}, \frac{(1-\theta-t_{01})p_{11}}{t_{01}}\right\} \\ &\quad - \max\left\{0, \frac{t_{01}(1-\theta-t_{01}-p_{00})}{1-\theta-t_{01}}\right\}. \end{aligned}$$

The fractional expressions apply when $0 < t_{01} < 1-\theta$. At $t_{01} = 0$, use the bounds $[p_{11}, 1-p_{10}]$ for $P(Y(1) = 1)$, and, at $t_{00} = 0$, use $[p_{01}, 1 - p_{00}]$ for $P(Y(0) = 1)$.

Let the feasible set of t_{01} be $\mathcal{T}_{01} = [\max\{0, P(A = 1) - \theta\}, \min\{1-\theta, P(A = 1)\}]$. Therefore, the general lower and upper bounds for both considered estimands, $(\mathcal{L}_{P(Y(A=1)=1)}, \mathcal{U}_{P(Y(A=1)=1)})$ and $(\mathcal{L}_{ATE}, \mathcal{U}_{ATE})$ are:

$$(\mathcal{L}_{P(Y(A=1)=1)}, \quad \mathcal{U}_{P(Y(A=1)=1)}) = (\min_{t_{01}\in\mathcal{T}_{01}} L_{P(Y(A=1)=1)}(t_{01}), \quad \max_{t_{01}\in\mathcal{T}_{01}} U_{P(Y(A=1)=1)}(t_{01})) \quad (7)$$

and

$$(\mathcal{L}_{ATE}, \quad \mathcal{U}_{ATE}) = (\min_{t_{01}\in\mathcal{T}_{01}} L_{ATE}(t_{01}), \quad \max_{t_{01}\in\mathcal{T}_{01}} U_{ATE}(t_{01})) \quad (8)$$

C.1.3 Sharpness for $P(Y(A=1)=1)$ and ATE bounds

Theorem 6. *The bounds calculated in Appendix C.1.2 are valid and sharp for* $P(Y(A=1)=1)$ *and the ATE.*

Proof. Validity follows from Appendix C.1.2. For sharpness, fix any feasible t_{01} and the resulting treatment masses t_{ha}. Independently for $a=0,1$, choose

$$q_{0a1} \in [\max\{0, t_{0a} - p_{a0}\}, \min\{t_{0a}, p_{a1}\}], \qquad q_{0a0} = t_{0a} - q_{0a1},$$

and set $q_{1ay} = p_{ay} - q_{0ay}$. Both intervals for q_{001} and q_{011} contain feasible values that determine probabilities that satisfy the observed data and sum to the fixed t_{ha}.

When $t_{00}, t_{01} > 0$, construct the unconfounded strata by

$$P(h_0 a_i y_{rs}) = t_{0i} \frac{q_{00r}}{t_{00}} \frac{q_{01s}}{t_{01}}, \qquad i, r, s \in \{0,1\}.$$

Those probabilities reproduce every q_{0ay}. Both treatment and control groups have the same joint potential outcome distribution in $H=0$, so all the unconfoundedness equations hold. Then, every pair (q_{001}, q_{011}) can be attained jointly, including opposite endpoints of their intervals. For $H=1$, each probability q_{1ay} can be split arbitrarily between the two strata with $Y(a)=y$, because the other potential outcome is unobserved.

Without loss of generality, consider $P(Y(1)=1)$, as the marginal bounds for $P(Y(0)=1)$ follow by relabeling treatment. For positive t_{00}, t_{01}, the construction gives

$$P(Y(1)=1) = p_{11} + \frac{t_{00}}{t_{01}} q_{011} + P(h_1 a_0 y_{01}) + P(h_1 a_0 y_{11}),$$

where $t_{00} = 1 - \theta - t_{01}$, $t_{10} = \theta + t_{01} - P(A=1)$, and $t_{11} = P(A=1) - t_{01}$. For the lower bound on $P(Y(1)=1)$, choose $q_{011} = \max\{0, t_{01} - p_{10}\}$ and put the untreated $H=1$ masses in $h_1 a_0 y_{00}$ and $h_1 a_0 y_{10}$, making their contribution zero. For the upper bound, choose $q_{011} = \min\{t_{01}, p_{11}\}$ and put those masses in $h_1 a_0 y_{01}$ and $h_1 a_0 y_{11}$, making their contribution

t_{10}. Thus the attained values are

$$\begin{aligned}
L_{P(Y(A=1)=1)}(t_{01}) &= p_{11} + \frac{t_{00}}{t_{01}} \max\{0, t_{01} - p_{10}\} \\
&= p_{11} + \max\left\{0, \frac{(1-\theta-t_{01})(t_{01}-p_{10})}{t_{01}}\right\}, \\
U_{P(Y(A=1)=1)}(t_{01}) &= p_{11} + \frac{t_{00}}{t_{01}} \min\{t_{01}, p_{11}\} + t_{10} \\
&= \theta + t_{01} - p_{10} + \min\left\{1-\theta-t_{01}, \frac{(1-\theta-t_{01})p_{11}}{t_{01}}\right\}.
\end{aligned}$$

To complete the strata distribution, choose any feasible q_{001}, which will not affect $P(Y(A = 1) = 1)$.

For the ATE, we also use

$$P(Y(0) = 1) = p_{01} + \frac{t_{01}}{t_{00}} q_{001} + P(h_1 a_1 y_{10}) + P(h_1 a_1 y_{11}).$$

For the lower ATE bound, choose $q_{011} = \max\{0, t_{01} - p_{10}\}$ and $q_{001} = \min\{t_{00}, p_{01}\}$. Within $H = 1$, assign $Y(1) = 0$ to untreated units and $Y(0) = 1$ to treated units, preserving their observed outcomes. Their missing-outcome contributions are then 0 and t_{11}, respectively. Subtracting the two expressions above gives

$$\begin{aligned}
L_{\text{ATE}}(t_{01}) &= p_{11} - p_{01} + \frac{t_{00}}{t_{01}} \max\{0, t_{01} - p_{10}\} \\
&\quad - \frac{t_{01}}{t_{00}} \min\{t_{00}, p_{01}\} - t_{11} \\
&= L_{P(Y(A=1)=1)}(t_{01}) - U_{P(Y(A=0)=1)}(t_{01}).
\end{aligned}$$

For the upper ATE bound, choose $q_{011} = \min\{t_{01}, p_{11}\}$ and $q_{001} = \max\{0, t_{00} - p_{00}\}$. Assign $Y(1) = 1$ to untreated $H = 1$ units and $Y(0) = 0$ to treated $H = 1$ units, giving contributions

t_{10} and 0, respectively. Consequently,

$$\begin{aligned} U_{\text{ATE}}(t_{01}) &= p_{11} - p_{01} + \frac{t_{00}}{t_{01}} \min\{t_{01}, p_{11}\} + t_{10} \\ &\quad - \frac{t_{01}}{t_{00}} \max\{0, t_{00} - p_{00}\} \\ &= U_{P(Y(A=1)=1)}(t_{01}) - L_{P(Y(A=0)=1)}(t_{01}). \end{aligned}$$

By this construction, both lower and upper bounds for the ATE are attained by distributions of strata that satisfy all the constraints, so proving sharpness.

It is relevant to consider the case where $t_{0a} = 0$. Without loss of generality, consider $t_{01} = 0$. As $P(Y(A = 1) = 1) = P(A = 1, Y = 1) + P(A = 0, Y(A = 1) = 1)$, if the unconfounded component contains no treated units, setting $Y(A = 1) = 0$ for all untreated units implies $P(A = 0, Y(A = 1) = 1) = 0$ and $P(Y(A = 1) = 1) = P(A = 1, Y = 1)$, attaining the lower bound. On the other hand, setting $Y(A = 1) = 1$ for all the untreated units implies $P(A = 0, Y(A = 1) = 1) = P(A = 0)$, so $P(Y(A = 1) = 1) = P(A = 1, Y = 1) + P(A = 0)$, attaining the upper bound. Each probability assignment satisfies the observed data by consistency and satisfies unconfoundedness within $H = 0$, as treatment is constant in that component. For ATE sharpness, at $t_{01} = 0$, setting $Y(A = 1) = 0$ for all untreated units and $Y(A = 0) = 1$ for all the treated units and, conversely, setting $Y(A = 1) = 1$ for all untreated units and $Y(A = 0) = 0$ for all the treated units will give respectively, $L_{ATE} = -p_{10} - p_{01}$ and $U_{ATE} = p_{11} + p_{00}$. This preserves observed data, and the ATE bounds are also sharp.

□

C.1.4 Optimizing over t_{01}

The bounds calculated in Equations 7 and 8 require searching over the feasible set of t_{01} of infinite cardinality. In fact, it is possible to reduce this search to finite sets. This is because possible maxima and minima can be found either in points of degeneracy or among the

extrema of feasible values of t_{01}. This can be shown for both $P(Y(A=1)=1)$ and for the ATE.

Theorem 7 (Evaluation of bounds for $P(Y(A=1)=1)$ at finite points of $\mathcal{T}_{01}$)**.** *Fix θ and use the fixed-allocation bounds derived above. The sharp bounds for $P(Y(A=1)=1)$ are*

$$\begin{aligned}\mathcal{L}_{P(Y(A=1)=1)}(\theta) = \min\Big\{&L_{P(Y(A=1)=1)}\big(\max\{0, P(A=1)-\theta\}\big),\\ &L_{P(Y(A=1)=1)}\big(\min\{1-\theta, P(A=1)\}\big)\Big\},\\ \mathcal{U}_{P(Y(A=1)=1)}(\theta) = \max\Big\{&U_{P(Y(A=1)=1)}\big(\max\{0, P(A=1)-\theta\}\big),\\ &U_{P(Y(A=1)=1)}\big(\min\{1-\theta, P(A=1)\}\big)\Big\}.\end{aligned}$$

Similarly, the sharp bounds for $P(Y(A=0)=1)$ are

$$\begin{aligned}\mathcal{L}_{P(Y(A=0)=1)}(\theta) = \min\Big\{&L_{P(Y(A=0)=1)}\big(\max\{0, P(A=1)-\theta\}\big),\\ &L_{P(Y(A=0)=1)}\big(\min\{1-\theta, P(A=1)\}\big)\Big\},\\ \mathcal{U}_{P(Y(A=0)=1)}(\theta) = \max\Big\{&U_{P(Y(A=0)=1)}\big(\max\{0, P(A=1)-\theta\}\big),\\ &U_{P(Y(A=0)=1)}\big(\min\{1-\theta, P(A=1)\}\big)\Big\}.\end{aligned}$$

All four fixed-allocation bounds use the same argument t_{01}, with $t_{00} = 1-\theta-t_{01}$. At any feasible allocation with $t_{01}=0$ or $t_{00}=0$, use

$$L_{P(Y(A=a)=1)}(t_{01}) = p_{a1}, \qquad U_{P(Y(A=a)=1)}(t_{01}) = 1-p_{a0}, \quad a\in\{0,1\}.$$

For $0 < t_{01} < 1-\theta$, use the fractional formulas derived above. Their branches change at p_{10} and p_{11} for $P(Y(A=1)=1)$, and at $1-\theta-p_{00}$ and $1-\theta-p_{01}$ for $P(Y(A=0)=1)$. Only points within the feasible interval are considered.

Proof. The key idea is to optimize the bounds derived above over the feasible interval $\mathcal{T}_{01} =$

$[t_{01}^-, t_{01}^+]$, where

$$t_{01}^- = \max\{0, P(A=1) - \theta\}, \qquad t_{01}^+ = \min\{1-\theta, P(A=1)\}.$$

First, if $0 \in \mathcal{T}_{01}$, then $L_{P(Y(A=1)=1)}(0) = p_{11}$, as no other bound can be smaller, as this is exactly the natural bound for confounded models. Second, if $0 \notin \mathcal{T}_{01}$, and if any feasible $t_{01} \leq p_{10}$, that will also include the minimum of the feasible set, and the lower bound is again p_{11}, which is the lowest possible value. At last, if $0 \notin \mathcal{T}_{01}$ and every feasible $t_{01} > p_{10}$, the second derivative of the lower bound with respect to t_{01} is nonpositive. Therefore, the function is concave, and the lower bound must be found at an endpoint of the feasible interval.

$$\begin{aligned} L_{P(Y(A=1)=1)}(t_{01}) &= 1 - \theta + P(A=1) - t_{01} - \frac{(1-\theta)p_{10}}{t_{01}}, \\ \frac{d^2}{dt_{01}^2} L_{P(Y(A=1)=1)}(t_{01}) &= -\frac{2(1-\theta)p_{10}}{t_{01}^3} \leq 0. \end{aligned}$$

Notice that this result works for computing lower bounds for $P(Y(A=1)=0), P(Y(A=0)=1), P(Y(A=0)=0)$, by changing the polarity of each binary A or Y.

The upper bound of $P(Y(A=1)=1)$ can be computed by maximizing $1 - P(Y(A=1)=0)$, which is equivalent to minimizing $P(Y(A=1)=0)$, so necessarily it requires checking the endpoints of $\mathcal{T}_{01}$.

□

Both $\mathcal{L}_{\text{ATE}}$ and $\mathcal{U}_{\text{ATE}}$ are constructed over differences of bounds of $P(Y(A=1)=1)$ and $P(Y(A=0)=1)$ evaluated for the whole feasible space of t_{01}. It remains to prove that the ATE bounds are attained at an endpoint of the feasible interval or at a feasible point where the formulas change.

Theorem 8 (Evaluation of bounds for the ATE at finite points of $\mathcal{T}_{01}$)**.** *Fix θ and combine*

the marginal bounds at the same allocation:

$$L_{\mathrm{ATE}}(t_{01}) = L_{P(Y(A=1)=1)}(t_{01}) - U_{P(Y(A=0)=1)}(t_{01}),$$
$$U_{\mathrm{ATE}}(t_{01}) = U_{P(Y(A=1)=1)}(t_{01}) - L_{P(Y(A=0)=1)}(t_{01}).$$

The sharp ATE bounds are

$$\begin{aligned}\mathcal{L}_{\mathrm{ATE}}(\theta) = \min\Big\{ & L_{\mathrm{ATE}}\big(\max\{0, P(A=1)-\theta\}\big),\\ & L_{\mathrm{ATE}}\big(\min\{1-\theta, P(A=1)\}\big),\\ & L_{\mathrm{ATE}}(p_{10}),\ L_{\mathrm{ATE}}(1-\theta-p_{01})\Big\},\\ \mathcal{U}_{\mathrm{ATE}}(\theta) = \max\Big\{ & U_{\mathrm{ATE}}\big(\max\{0, P(A=1)-\theta\}\big),\\ & U_{\mathrm{ATE}}\big(\min\{1-\theta, P(A=1)\}\big),\\ & U_{\mathrm{ATE}}(p_{11}),\ U_{\mathrm{ATE}}(1-\theta-p_{00})\Big\}.\end{aligned}$$

In each ATE formula, include either of the last two evaluations only if its argument belongs to $[\max\{0, P(A=1)-\theta\}, \min\{1-\theta, P(A=1)\}]$. *At a feasible zero-mass allocation, the marginal conventions give* $L_{\mathrm{ATE}}(t_{01}) = -p_{10} - p_{01}$ *and* $U_{\mathrm{ATE}}(t_{01}) = p_{11} + p_{00}$.

Proof. First, notice that for feasible $t_{01} = 0$ or $t_{01} = 1-\theta$, then the results in Theorem 7 are sufficient.

Let $0 < t_{01} < 1-\theta$. If $t_{01} \leq p_{10}$, then the max inside the lower bound formula is 0. Also, for the min formula, if $t_{01} \leq \frac{p_{01}t_{01}}{1-\theta-t_{01}}$, which is equivalent to $t_{01} \geq 1-\theta-p_{01}$, then the min formula is equal to t_{01}. When those two conditions hold, the lower bound is constant for any t_{01}. For the other cases, it is sufficient to check the second derivative of the other nonaffine terms, which are equal to:

$$-\frac{2(1-\theta)p_{10}}{t_{01}^3}, \qquad -\frac{2(1-\theta)p_{01}}{(1-\theta-t_{01})^3}.$$

As both are nonpositive, the lower bound is concave on each piece. Its minimum therefore occurs at a feasible interval endpoint or at one of these two points, when feasible.

For the upper ATE bound, the formulas change at $t_{01} = p_{11}$ and $t_{01} = 1 - \theta - p_{00}$. The corresponding second derivatives are

$$\frac{2(1-\theta)p_{11}}{t_{01}^3}, \qquad \frac{2(1-\theta)p_{00}}{(1-\theta-t_{01})^3}.$$

Thus the upper bound is convex on each piece, and its maximum occurs at an interval endpoint or at one of these two points. That completes the proof.

□

C.1.5 Monotonicity of the bounds

Theorem 9. *For both $P(Y(1) = 1)$ and the ATE, the sharp lower bound is nonincreasing in θ, and the sharp upper bound is nondecreasing.*

Proof. Let $0 \leq \theta < \theta' \leq 1$ and any feasible strata distribution P at θ. Construct a distribution at θ' by moving the same fraction of every $H = 0$ stratum into its corresponding $H = 1$ stratum:

$$\begin{aligned} P'(h_0 a_i y_{rs}) &= \frac{1-\theta'}{1-\theta} P(h_0 a_i y_{rs}), \\ P'(h_1 a_i y_{rs}) &= P(h_1 a_i y_{rs}) + \frac{\theta' - \theta}{1-\theta} P(h_0 a_i y_{rs}). \end{aligned}$$

These probability masses are nonnegative, sum to one, and satisfy $P'(H = 1) = \theta'$. For $\theta' < 1$, the conditional on $H = 0$ strata distribution, i.e. the strata divided by $1 - \theta'$, remains unchanged, so the independence between treatment and potential outcomes is preserved. On the other hand, for $\theta' = 1$, the conditional on $H = 0$ strata distribution is undefined, because no unconfounded component exists anymore. Moreover, $\sum_{h=0}^{1} P'(h_h a_i y_{rs}) = \sum_{h=0}^{1} P(h_h a_i y_{rs})$. Therefore, both quantities $P(A = a, Y(A = 0) = y, Y(A = 1) = y')$ and,

when $\theta' < 1$, $P(A = a, Y(A = 0) = y, Y(A = 1) = y'|H = 0)$ are invariant.

The key idea is to show that allowing more confounding cannot rule out an effect value that was previously possible. So, if both optimization problems impose the same constraints with the exception of the probability mass of the confounded components, which increases from θ to θ', that suffices to prove the claim, provided that the construction preserves the effect value.

Let P be a strata distribution attaining the sharp lower bound $L(\theta)$. Then let the distributions be preserved as above. By increasing θ to θ', then the constructed distribution is feasible at θ' and still attains $L(\theta)$, so the sharp bound $L(\theta') \leq L(\theta)$. Similarly, applying the construction to a distribution attaining the sharp $U(\theta)$ will imply $U(\theta') \geq U(\theta)$.

□

C.2 Symbolic sensitivity functions in instrumental variable models

C.2.1 Polynomial Program for "no defiers" and exclusion restriction violations

Let $C = \{A(z_1) > A(z_0)\}$ denote the complier event. First, consider violations of "no defiers", retaining exogeneity and exclusion. Write $a_{rs}y_{tu}$ for the stratum satisfying

$$(A(z_0), A(z_1)) = (r, s), \qquad (Y(a_0), Y(a_1)) = (t, u),$$

where all indices are binary. By consistency and exogeneity, the observed cells are

$$\begin{aligned}
\Pr(y_0, a_0 | z_0) &= P(a_{00}y_{00}) + P(a_{01}y_{00}) + P(a_{00}y_{01}) + P(a_{01}y_{01}),\\
\Pr(y_0, a_0 | z_1) &= P(a_{00}y_{00}) + P(a_{10}y_{00}) + P(a_{00}y_{01}) + P(a_{10}y_{01}),\\
\Pr(y_0, a_1 | z_0) &= P(a_{11}y_{00}) + P(a_{10}y_{00}) + P(a_{11}y_{10}) + P(a_{10}y_{10}),\\
\Pr(y_0, a_1 | z_1) &= P(a_{11}y_{00}) + P(a_{01}y_{00}) + P(a_{11}y_{10}) + P(a_{01}y_{10}),\\
\Pr(y_1, a_0 | z_0) &= P(a_{00}y_{11}) + P(a_{01}y_{11}) + P(a_{00}y_{10}) + P(a_{01}y_{10}),\\
\Pr(y_1, a_0 | z_1) &= P(a_{00}y_{11}) + P(a_{10}y_{11}) + P(a_{00}y_{10}) + P(a_{10}y_{10}),\\
\Pr(y_1, a_1 | z_0) &= P(a_{11}y_{11}) + P(a_{10}y_{11}) + P(a_{11}y_{01}) + P(a_{10}y_{01}),\\
\Pr(y_1, a_1 | z_1) &= P(a_{11}y_{11}) + P(a_{01}y_{11}) + P(a_{11}y_{01}) + P(a_{01}y_{01}),
\end{aligned}$$

The stratum probabilities satisfy

$$P(a_{rs}y_{tu}) \geq 0, \qquad \sum_{r,s,t,u} P(a_{rs}y_{tu}) = 1, \qquad \sum_{t,u} P(a_{10}y_{tu}) = \theta_{\text{ND}}.$$

The last equality fixes the proportion of defiers. The objective is

$$\text{LATE} = \frac{P(a_{01}y_{01}) - P(a_{01}y_{10})}{\sum_{t,u} P(a_{01}y_{tu})}. \tag{9}$$

Minimizing and maximizing this expression over the stratum probabilities, subject to the observed-cell equations and the three restrictions above, defines the perturbation problem. Its denominator is identified as

$$P(C) = \sum_{t,u} P(a_{01}y_{tu}) = P(A = 1 \mid Z = 1) - P(A = 1 \mid Z = 0) + \theta_{\text{ND}} > 0.$$

Thus we can minimize and maximize the linear numerator and divide its bounds by $P(C)$. For the ATE, the same constraints apply to the objective $\sum_{r,s}[P(a_{rs}y_{01}) - P(a_{rs}y_{10})]$. Replacing the fixed-defier equality by $\sum_{t,u} P(a_{10}y_{tu}) \leq \theta_{\text{ND}}^{\max}$ gives the sensitivity problem; its LATE denominator then varies with the defier mass.

Next, consider violations of exclusion, retaining exogeneity and monotonicity. Write $a_{ij}y_{rstu}$ for the stratum satisfying

$$(A(z_0), A(z_1)) = (i, j), \qquad \big(Y(z_0, a_0), Y(z_0, a_1), Y(z_1, a_0), Y(z_1, a_1)\big) = (r, s, t, u).$$

All indices are binary. Monotonicity excludes the a_{10} strata. By consistency and exogeneity, the observed cells are, for $y = 0, 1$,

$$\begin{aligned}
P(Y = y, A = 0 \mid Z = 0) &= \sum_{s,t,u} \big[P(a_{00}y_{ystu}) + P(a_{01}y_{ystu})\big], \\
P(Y = y, A = 1 \mid Z = 0) &= \sum_{r,t,u} P(a_{11}y_{rytu}), \\
P(Y = y, A = 0 \mid Z = 1) &= \sum_{r,s,u} P(a_{00}y_{rsyu}), \\
P(Y = y, A = 1 \mid Z = 1) &= \sum_{r,s,t} \big[P(a_{01}y_{rsty}) + P(a_{11}y_{rsty})\big].
\end{aligned}$$

The stratum probabilities satisfy

$$P(a_{ij}y_{rstu}) \geq 0, \qquad \sum_{i,j,r,s,t,u} P(a_{ij}y_{rstu}) = 1, \qquad P(a_{10}y_{rstu}) = 0 \quad \text{for every } r, s, t, u.$$

Exclusion fails when $r \neq t$ or $s \neq u$, so its exact violation proportion imposes the additional constraint

$$\sum_{i,j} \sum_{\substack{r,s,t,u: \\ r \neq t \text{ or } s \neq u}} P(a_{ij}y_{rstu}) = \theta_{\text{ER}}.$$

The two complier numerators are

$$N_0 = \sum_{r,s,t,u} (s - r) P(a_{01} y_{rstu}), \qquad N_1 = \sum_{r,s,t,u} (u - t) P(a_{01} y_{rstu}).$$

The objective function, after averaging over the observed distribution over Z, is

$$\text{LATE} = \frac{P(z_0) N_0 + P(z_1) N_1}{\sum_{r,s,t,u} P(a_{01} y_{rstu})}. \tag{10}$$

Minimizing and maximizing this expression over the stratum probabilities, subject to the observed-cell equations, probability constraints, monotonicity, and the specified proportion of exclusion violations, defines the perturbation problem. Here its denominator is identified as

$$P(C) = \sum_{r,s,t,u} P(a_{01} y_{rstu}) = P(A = 1 \mid Z = 1) - P(A = 1 \mid Z = 0) > 0.$$

Consequently, optimizing the weighted numerator jointly and dividing by $P(C)$ gives sharp bounds. Replacing the fixed-exclusion equality by an upper bound $\theta_{\text{ER}}^{\max}$ gives the corresponding sensitivity program.

C.2.2 Proof of the LATE bounds under monotonicity violations and under exclusion violations

Theorem 2 (Perturbation bounds on the LATE for monotonicity violations)**.** *Let $C = \{A(z_1) > A(z_0)\}$ denote the complier event. Suppose exogeneity and exclusion hold, the observed distribution is compatible with the model and a fixed specified θ_{ND}, and $P(C) = P(A(z_1) > A(z_0)) > 0$. Then the perturbation bounds for a particular θ_{ND} are $l(\theta_{ND})/P(C) \leq LATE \leq u(\theta_{ND})/P(C)$, where $p_{ij}^k = P(Y = y_i, A = a_j \mid Z = z_k)$ for $i, j, k \in \{0, 1\}$. The*

functions $l(\theta_{\mathrm{ND}})$ *and* $u(\theta_{\mathrm{ND}})$ *are, respectively,*

$$\max\begin{cases} -p_{10}^0, \\ p_{11}^1 - P(y_1|z_0), \\ p_{11}^1 + p_{00}^0 - 1 + \theta_{ND}, \\ -P(C), \\ p_{10}^1 - p_{10}^0 - \theta_{ND}, \\ p_{01}^0 - p_{01}^1 - \theta_{ND}, \\ P(y_0|z_0) - P(y_0|z_1) - \theta_{ND}, \\ -p_{01}^1, \\ p_{00}^0 - P(y_0|z_1), \end{cases} , \quad \min\begin{cases} 1 - p_{10}^0 - p_{01}^1 - \theta_{ND}, \\ p_{11}^1, \\ P(y_0|z_0) - p_{01}^1, \\ P(y_0|z_0) - P(y_0|z_1) + \theta_{ND}, \\ p_{00}^0, \\ P(C), \\ p_{00}^0 - p_{00}^1 + \theta_{ND}, \\ P(y_1|z_1) - p_{10}^0, \\ p_{11}^1 - p_{11}^0 + \theta_{\mathrm{ND}}, \end{cases}$$

and $P(C) = P(a_1|z_1) - P(a_1|z_0) + \theta_{ND}$.

Theorem 3 (Perturbation bounds on the LATE under exclusion violations)**.** *Here, with exclusion relaxed, write* $\mathrm{LATE} = \mathbb{E}[Y(Z,1) - Y(Z,0) \mid A(z_1) > A(z_0)]$*, which reduces to the usual LATE under exclusion. Suppose exogeneity and monotonicity hold, the observed distribution is compatible with the model and the specified* θ_{ER}*, and* $P(C) = P(A(z_1) > A(z_0)) > 0$*. Then* LATE *satisfies* $l(\theta_{\mathrm{ER}})/P(C) \leq \mathrm{LATE} \leq u(\theta_{\mathrm{ER}})/P(C)$*, using* p_{ij}^k *as above.*
The functions $l(\theta_{\mathrm{ER}})$ *and* $u(\theta_{\mathrm{ER}})$ *are, respectively,*

$$\sum_{k=0}^{1} P(z_k) \max\begin{cases} -P(C), \\ -p_{1-k,k}^k, \\ P(y_1|z_1) - P(y_1|z_0) - \theta_{\mathrm{ER}}, \\ p_{01}^0 - p_{00}^0 + p_{00}^1 - p_{01}^1 - \theta_{\mathrm{ER}}, \\ p_{10}^1 - p_{11}^1 - p_{10}^0 + p_{11}^0 - \theta_{\mathrm{ER}}, \end{cases} , \sum_{k=0}^{1} P(z_k) \min\begin{cases} P(C), \\ p_{kk}^k, \\ P(y_1|z_1) - P(y_1|z_0) + \theta_{\mathrm{ER}}, \\ p_{00}^0 - p_{01}^0 - p_{00}^1 + p_{01}^1 + \theta_{\mathrm{ER}}, \\ p_{10}^0 - p_{11}^0 + p_{11}^1 - p_{10}^1 + \theta_{\mathrm{ER}}. \end{cases}$$

with $P(C) = P(a_1|z_1) - P(a_1|z_0)$.

Let $p_{z1} = P(Z = 1)$*,* $\min_z = \min\{p_{z1}, 1 - p_{z1}\}$*, and* $\max_z = \max\{p_{z1}, 1 - p_{z1}\}$*. Define*

$$a_{p_{z1}} = \begin{cases} p_{10}^0, & p_{z1} \leq \frac{1}{2}, \\ p_{01}^1, & p_{z1} > \frac{1}{2}, \end{cases} \qquad b_{p_{z1}} = \begin{cases} p_{00}^0, & p_{z1} \leq \frac{1}{2}, \\ p_{11}^1, & p_{z1} > \frac{1}{2}. \end{cases}$$

For $\theta = \theta_{\text{ER}}$, the sharp perturbation bounds are

$$L_{\text{ER}}^{\text{pert}}(\theta) = \frac{1}{P(C)} \max\Big\{l(\theta),\ -\max_z P(C) + \min_z(\theta - 1),\\ -\min_z P(C) - |1 - 2p_{z1}|a_{p_{z1}} + \min_z(\theta - 1)\Big\},$$

$$U_{\text{ER}}^{\text{pert}}(\theta) = \frac{1}{P(C)} \min\Big\{u(\theta),\ \max_z P(C) + \min_z(1 - \theta),\\ \min_z P(C) + |1 - 2p_{z1}|b_{p_{z1}} + \min_z(1 - \theta)\Big\}.$$

Proof of Theorems 2 and 3. First, notice that the denominator is point identifiable for both problems. Under the monotonicity relaxation case, after fixing $d = \theta_{\text{ND}}$, then $P(C) = P(A = 1 \mid Z = 1) - P(A = 1 \mid Z = 0) + d$. Under the exclusion restriction relaxation case, $P(C) = P(A = 1 \mid Z = 1) - P(A = 1 \mid Z = 0)$. So, it is sufficient to calculate bounds for the numerator. The problem becomes solving both linear programs defined in Appendix C.2.1.

The linear programs can be solved symbolically using the Python interface to `cddlib`, with exact rational arithmetic, which is employed in the `SymbolicProblem` class in `autobounds`. Appendix E provides minimal standalone code for both perturbation calculations that makes use of `cddlib` and `SymPy` but without the `autobounds` framework. As solutions for a linear program, the calculated bounds are sharp. The final stated bounds are divisions of those bounds by the identified denominator, assuming $P(C) > 0$.

For the exclusion restriction perturbation function, solve the program for the maximum and minimum, where the objective function is $N = (1 - p_{z1})N_0 + p_{z1}N_1$, over probability distributions on the 48 principal strata, excluding the ones that do not satisfy monotonicity. The program must include the constraints of observed data and the exact violation proportion θ_{ER}. The solution is obtained by enumerating vertices, using exact rational projection in `cddlib`. This yields the lower and upper numerator envelopes displayed in Theorem 3. The feasible set is a nonempty compact polytope, so both extrema are attained, which gives the sharpness of the bounds on the numerator. As $P(C) > 0$ is identified, then division by it gives the stated sharp perturbation bounds. □

C.2.3 From IV perturbation bounds to sensitivity bounds

Theorem 4 (Natural sensitivity bounds for IV assumption violations)**.** *Consider either relaxation separately, under the maintained assumptions of Theorems 2 and 3, respectively, and suppose its sensitivity model is nonempty. Write* $q_a^z = P(A = a \mid Z = z)$ *and let* C *denote the complier event.*

(i) For monotonicity sensitivity, let $\mathcal{D}$ *contain the feasible values* $d \leq \theta_{\text{ND}}^{\max}$ *with* $P(C) = q_1^1 - q_1^0 + d > 0$*. Let* $\mathcal{B}$ *consist of the endpoints of* $\mathcal{D}$ *and the values of* d *within* $\mathcal{D}$ *at which two nonparallel affine terms in either* $l(d)$ *or* $u(d)$ *intersect. The sharp LATE sensitivity endpoints are*

$$L_{\text{ND}}(\theta_{\text{ND}}^{\max}) = \min_{d \in \mathcal{B}} \frac{l(d)}{q_1^1 - q_1^0 + d}, \qquad U_{\text{ND}}(\theta_{\text{ND}}^{\max}) = \max_{d \in \mathcal{B}} \frac{u(d)}{q_1^1 - q_1^0 + d},$$

using one-sided limits at excluded endpoints.

(ii) For exclusion sensitivity, $P(C) = q_1^1 - q_1^0 > 0$ *is fixed and the sensitivity envelopes of the displayed bounds for* LATE *are*

$$L_{\text{ER}}(\theta_{\text{ER}}^{\max}) = \frac{l(\theta_{\text{ER}}^{\max})}{P(C)}, \qquad U_{\text{ER}}(\theta_{\text{ER}}^{\max}) = \frac{u(\theta_{\text{ER}}^{\max})}{P(C)}.$$

These sensitivity bounds are sharp.

Proof of Theorem 4. We start with the proof for monotonicity sensitivity.

For the proof, consider the lower bound of the sensitivity function. The proof for the upper bounds follows analogously.

Denote the probability of defiers by d. The sensitivity lower bound is

$$\min_{\substack{d \text{ feasible} \\ d \leq \theta^{\max}}} L_{\text{pert}}(d).$$

The proof reduces this potentially infinite set of candidates d to a finite set.

The key idea of this proof is for each expression of each perturbation function, one can calculate endpoints and switching points for a feasible interval for d. Then for each candidate d, one can evaluate the maximum of the lower bound numerator expression and divide them by $P(C)$. Finally, among the many candidates of d, take the minimum of the resulting lower bound values.

Write $p^z_{ya} = P(Y = y, A = a \mid Z = z)$ and $q^z_a = P(A = a \mid Z = z)$. With defier proportion d, the complier probability is $P(C) = q^1_1 - q^0_1 + d$. Under exogeneity and exclusion, the feasible defier interval is

$$d_- = \max\{0, q^1_0 + q^0_1 - 1, p^1_{00} - p^0_{00}, p^1_{10} - p^0_{10}, p^0_{01} - p^1_{01}, p^0_{11} - p^1_{11}\},$$
$$d_+ = \min\{q^1_0, q^0_1\}.$$

The upper bound can first be calculated by noticing that the observed cells $A = 0$ under $Z = 1$ and $A = 1$ under $Z = 0$ each contain all the defier strata $a_{10}y_{00}, a_{10}y_{01}, a_{10}y_{10}, a_{10}y_{11}$. The lower bound can be calculated by covering all the other 12 strata with observed cells and taking the smallest sum of their probabilities. One minus this upper bound on the probability of all nondefier strata will be the lower bound for the defiers. Fixing any $d \in [d_-, d_+]$, then the probability of never-takers, always-takers, and compliers are respectively identified by $q^1_0 - d$, $q^0_1 - d$, $q^1_1 - q^0_1 + d$. Then, for each of those values, the mass can be distributed, matching each observed cell, across the outcome y-strata: $y_{00}, y_{01}, y_{10}, y_{11}$.

Let $\mathcal{D} = [d_-, d_+] \cap [0, \theta^{\max}_{\mathrm{ND}}]$. Between intersections of the displayed affine terms, each LATE endpoint has the form $(a + bd)/(q^1_1 - q^0_1 + d)$, with $P(C) = q^1_1 - q^0_1 + d > 0$. Its first derivative with respect to d is

$$\frac{b(q^1_1 - q^0_1) - a}{(q^1_1 - q^0_1 + d)^2}.$$

As the sign is constant, then the extrema over $\mathcal{D}$ occur among the stated endpoints or switching points, so there is no need to search for interior ones.

For exclusion sensitivity, impose the constraint

$$P\{\exists a : Y(0,a) \neq Y(1,a)\} \leq \theta_{\text{ER}}^{\max}.$$

The function $l(\theta)$ is nonincreasing and $u(\theta)$ is nondecreasing, so their values at $\theta_{\text{ER}}^{\max}$ give valid bounds for the numerator. Then minimizing and maximizing $(1-p_{z1})N_0 + p_{z1}N_1$ over the 48 principal strata not excluded by the monotonicity assumption, under the constraints of observed data and maximum violation of $\theta_{\text{ER}}^{\max}$, gives $l(\theta_{\text{ER}}^{\max})$ and $u(\theta_{\text{ER}}^{\max})$ by exact symbolic optimization. Both extrema are attained on the nonempty compact feasible polytope, which proves sharpness. Dividing by the identified $P(C) > 0$ gives the sharp sensitivity bounds. □

D Statistical Inference

D.1 Endpoint estimation and confidence limits

So far, sharp sensitivity curves capture only structural uncertainty, which remains even if the population distribution of the observed variables is known. With sample data, the estimated endpoints also contain sampling uncertainty. Fix an allowed violation vector $\Theta^{\max}$ and suppose that the data are independent and identically distributed. Let $L = L(\Theta^{\max})$ and $U = U(\Theta^{\max})$ denote the population endpoints, and let $[\widehat{L}_n, \widehat{U}_n]$ be their plug-in estimates obtained from the empirical distribution. Notice that the goal is to estimate the curves, i.e. the lower and upper bounds of the partially identified region, not the region itself, which might contain holes. The confidence intervals here concern the endpoints. By contrast, Imbens and Manski (2004) construct confidence intervals for the true parameter rather than the entire identified region.

Inference below is pointwise at the fixed sensitivity vector $\Theta^{\max}$. Consistency of these estimators requires regularity conditions on the optimization problem. In particular, suppose that its feasible sets are nonempty and compact in a neighborhood of the population

distribution and that the endpoint value functions are continuous there. The continuous mapping theorem then implies $(\widehat{L}_n, \widehat{U}_n) \xrightarrow{p} (L, U)$. Proposition 5 proves this fact. A harder problem is if those estimators converge in distribution and to which distribution. It is not possible to conclude that those estimators converge at root-n rates to Gaussian distributions, because maxima and minima are in general only directionally differentiable. This lack of differentiability means that naive applications of bootstrap or standard delta method might fail (Andrews and Han, 2009), although specialized versions of both methods can work (Bugni, 2010; Fang and Santos, 2019).

The key result shown here is that confidence intervals for the bounds, given some $\theta^{\max}$, can be estimated using recentered subsampling (Romano and Shaikh, 2010; Cooper et al., 2026). The validity of this method depends on the conditions that the estimators of the curves are consistent and

$$\sqrt{n}\begin{pmatrix}\widehat{L}_n - L\\ \widehat{U}_n - U\end{pmatrix} \rightsquigarrow \begin{pmatrix}Z_L\\ Z_U\end{pmatrix},$$

where the marginal distribution functions are continuous and strictly cross the quantile levels used for inference, as specified in Proposition 7. These limits do not need to be normally distributed. For bounds expressed as finite maxima or minima of smooth functions, Proposition 6 below establishes consistency and the limiting distribution directly from the central limit theorem. Continuity of the limiting distribution and the quantile conditions still require attention in degenerate cases. This argument does not establish the same conclusions for arbitrary polynomial programs. Proposition 7 proves that subsampling consistently estimates their marginal distributions when $m \to \infty$ and $m/n \to 0$ (Politis et al., 2001).

This is the procedure. Set $m = \lfloor n^\gamma \rfloor$, where $0 < \gamma < 1$. The empirical applications use $\gamma = 0.8$. Draw B independent subsets of size m, uniformly and without replacement within each subset, recompute $[\widehat{L}_m^{(b)}, \widehat{U}_m^{(b)}]$ in each subsample, and form $T_L^{(b)} = \sqrt{m}\big(\widehat{L}_m^{(b)} - \widehat{L}_n\big)$ and $T_U^{(b)} = \sqrt{m}\big(\widehat{U}_m^{(b)} - \widehat{U}_n\big)$. This choice of m satisfies the two rate conditions. If $\widehat{q}_{L,p}$ and $\widehat{q}_{U,p}$ denote the empirical p-quantiles of these statistics, respectively, the one-sided confidence

limits are

$$\ell_n(\theta) = \widehat{L}_n(\theta) - \frac{\widehat{q}_{L,1-\alpha}(\theta)}{\sqrt{n}}, \qquad u_n(\theta) = \widehat{U}_n(\theta) - \frac{\widehat{q}_{U,\alpha}(\theta)}{\sqrt{n}}.$$

At each fixed sensitivity value, ℓ_n is a lower confidence limit for L, and u_n is an upper confidence limit for U, each at asymptotic level $1-\alpha$. It is important to emphasize one more time, those are not a joint $(1-\alpha)$ confidence interval for the identified set, but separate one-sided statements about the bounds.

Besides sensitivity curves, inference on robustness values has specific challenges. In the first place, this in population terms represents the amount of relaxation a conclusion about an estimand sign can withstand. The concern is typically that true robustness is smaller than estimated. Assume the initial estimand is positive. As larger robustness does not undermine this guarantee, one-sided lower confidence intervals are sufficient. This is how it is calculated. For a prespecified positive effect direction, assume the population bounds are attained, $L(0) > 0$, and L is continuous and nonincreasing. Define

$$\underline{\theta}_n = \inf\big(\{\theta \in [0,1] : \ell_n(\theta) \leq 0\} \cup \{1\}\big).$$

Then $P(\underline{\theta}_n \leq \theta^\star)$ has limit inferior at least $1-\alpha$: coverage at an interior true crossing implies that the lower confidence limit has reached zero no later. If there is no crossing, however, $\theta^\star = 1$, the inequality holds true automatically, and $P(\underline{\theta}_n \leq \theta^\star) = 1 \geq 1-\alpha$. If the estimand is initially negative, then it is sufficient to replace ℓ_n by $-u_n$. As no upper confidence bound on robustness is sought, subsamples whose curves never cross are not discarded. If the robustness value is evaluated by grid search, on a fixed grid, report the grid point immediately before the first point where the lower confidence limit is nonpositive. Report zero if it is nonpositive at the first point, and one if it remains positive throughout the grid. Rounding down preserves the lower-confidence-bound guarantee when the true threshold lies between grid points. The proof is in Appendix D.3.

D.2 Proofs of consistency and subsampling validity

The following results concern consistency and pointwise coverage of the estimated sensitivity bounds given a fixed relaxation parameter under the population distribution. Therefore, they do not target the simultaneous coverage of the whole sensitivity curve.

Proposition 5 (Consistency of estimated endpoints)**.** *Let p be the vector of observed cell probabilities in a finite model, and let $\widehat{p}_n$ be its empirical estimate from independent, identically distributed observations. Fix the sensitivity parameter. Suppose the feasible sets are nonempty and compact on a neighborhood of p in the probability simplex, the objective is defined and continuous on those sets, and the endpoint functions $L(p)$ and $U(p)$ are continuous at p. Then:*

$$\big(L(\widehat{p}_n), U(\widehat{p}_n)\big) \xrightarrow{p} \big(L(p), U(p)\big).$$

Proof. The law of large numbers gives $\widehat{p}_n \to p$ in probability, and in consequence, $\widehat{p}_n$ is consistent for p. The polynomial programs are constructed over closed subsets of the probability simplex. Assume all p probabilities are positive and the bound functions are defined near p and continuous at it. By the continuous mapping theorem, therefore, the bounds are also consistent. For cases where some p cells are equal to 0, then their empirical frequencies are also zero almost surely. Then the empirical $\widehat{p}_n$ stays in subset of the simplex where those cells have probability zero, and continuity outside that set is irrelevant. If the bound functions are defined near p within this subset and continuous at p along it, then the empirical bounds are again consistent. □

Proposition 6 (Limits for finite maxima and minima)**.** *Fix the sensitivity parameter. Let p be the vector of observed cell probabilities and suppose $\sqrt{n}(\widehat{p}_n - p) \rightsquigarrow G$, where G is a centered Gaussian vector. Suppose an endpoint is $T(p) = \max_{1 \le j \le J} f_j(p)$, where J is finite and each f_j is continuously differentiable in a neighborhood of p. Write $I(p) = \{j : f_j(p) = T(p)\}$.*

Then $T(\widehat{p}_n)$ is consistent and

$$\sqrt{n}\{T(\widehat{p}_n) - T(p)\} \rightsquigarrow \max_{j \in I(p)} \nabla f_j(p)^\top G.$$

For a minimum, both maxima in this statement are replaced by minima.

Proof. Consistency of $T(\widehat{p}_n)$ follows from Proposition 5. Any function f_j that is inactive, i.e. is not a maximum, has a strictly positive gap below the maximum at p. The fact that the number of functions f_j is finite and that they are continuous implies that, with probability tending to one, only the indices in $I(p)$ can attain the empirical maximum asymptotically.

For each index in $I(p)$, Taylor expansion gives

$$\sqrt{n}\{f_j(\widehat{p}_n) - f_j(p)\} = \nabla f_j(p)^\top \sqrt{n}(\widehat{p}_n - p) + o_p(1).$$

The remainders are jointly $o_p(1)$ because there are finitely many functions. Taking their maximum and applying the continuous mapping theorem proves the result.

This proof also works for the minimum by the fact that $\min_j f_j = -\max_j(-f_j)$. □

The result in Proposition 6 holds for multinomial proportions whose Gaussian limit has covariance $\mathrm{diag}(p) - pp^\top$, conditional cell probabilities with positive conditional probabilities, and finite sums and nested maxima or minima of smooth expressions. For ratios, such as LATE, the result holds at a fixed population distribution as long as the denominator is strictly positive.

The limiting distribution need not be continuous. If a bound is constant in a neighborhood of the population distribution, its estimate equals that constant with probability tending to one. In this case, the subsampling confidence limit has coverage tending to one. Otherwise, the continuity and quantile conditions in the following proposition still need to be verified.

A separate issue is degeneracy of the limit. For a single finite maximum or minimum,

positive variance of every active Gaussian term is sufficient for a continuous limiting distribution: an atom in their maximum or minimum would require an atom in at least one term. A locally constant bound, for example, instead has a degenerate limit. Thus a positive LATE denominator alone does not establish all the distributional conditions of the next proposition. Whenever its continuity and quantile conditions hold, recentered subsampling estimates the limit derived above, including at ties between active smooth expressions.

Proposition 7 (Recentered subsampling and one-sided coverage)**.** *For either endpoint $T \in \{L, U\}$ at a fixed sensitivity value, let $\widehat{T}_n$ be a measurable, permutation-invariant statistic of an i.i.d. sample. Suppose*

$$\sqrt{n}(\widehat{T}_n - T) \rightsquigarrow Z_T,$$

where Z_T has continuous distribution function F_T. Let $m = m_n \to \infty$, $m/n \to 0$, and $B = B_n \to \infty$. Conditional on the data, draw B independent, uniformly selected subsets of m distinct observations and recompute the same statistic on each. Define

$$\widehat{F}_{n,m,B}(x) = \frac{1}{B} \sum_{b=1}^{B} \mathbb{1}\{\sqrt{m}(\widehat{T}_m^{(b)} - \widehat{T}_n) \leq x\}.$$

Then $\sup_x |\widehat{F}_{n,m,B}(x) - F_T(x)| \to 0$ in probability. For a quantile level $v \in (0,1)$, additionally assume that $q_{T,v} = \inf\{x : F_T(x) \geq v\}$ satisfies

$$F_T(q_{T,v} - \varepsilon) < v < F_T(q_{T,v} + \varepsilon) \quad \text{for every } \varepsilon > 0.$$

Its subsampling estimate $\widehat{q}_{T,v}$ then converges in probability to $q_{T,v}$. Applying this condition at $v = 1 - \alpha$ for L and $v = \alpha$ for U gives

$$P\left(\widehat{L}_n - \frac{\widehat{q}_{L,1-\alpha}}{\sqrt{n}} \leq L\right) \longrightarrow 1 - \alpha, \qquad P\left(U \leq \widehat{U}_n - \frac{\widehat{q}_{U,\alpha}}{\sqrt{n}}\right) \longrightarrow 1 - \alpha.$$

Alternatively, suppose the endpoint function is constant in a neighborhood of the population distribution. In the locally constant case, the relevant subsampling quantiles converge to zero, and the one-sided confidence limit has coverage tending to one. Then

$$\liminf_{n\to\infty} P(\textit{confidence limit covers the endpoint}) \geq 1-\alpha.$$

A sufficient condition for incompatibility to vanish is that the true observed distribution lies in the interior of the compatible region, relative to its sampling support. At binding compatibility boundaries, incompatibility may persist. The fallback values assigned to infeasible samples do not by themselves establish the weak-limit assumptions above, so the stated coverage result does not automatically extend to these cases.

Proof. For the distribution formed using all subsets, uniform convergence and one-sided coverage follow from Theorem 2.1(iii)–(iv) of Politis et al. (2001). In their notation, take $j_n = k_n = m_n$ and $\tau_n = \sqrt{n}$. The required conditions follow from $m \to \infty$ and $m/n \to 0$, since $\tau_m/\tau_n = \sqrt{m/n} \to 0$. The assumed weak limit and continuity of F_T provide their distributional assumptions. The same conclusions hold for the Monte Carlo approximation as $B \to \infty$. The stated strict crossing condition gives quantile consistency and therefore both coverage statements.

For the locally constant case, the plug-in estimate equals the true constant with probability tending to one, and the fraction of subsample estimates differing from that constant tends to zero. Then each relevant subsampling quantile is exactly zero with probability tending to one, giving coverage tending to one. □

D.3 Coverage of the one-sided robustness bound

Proposition 8 (Lower robustness bound)**.** *Suppose the identified sets are nonempty and expanding as θ increases. Suppose their bounds are attained and L is continuous on $[0,1]$. Assume $L(0) > 0$. Let $\theta^\star = \inf(\{\theta \in [0,1] : L(\theta) \leq 0\} \cup \{1\})$ denote the population robustness*

value.

Let $\ell_n(\theta)$ be the lower confidence limit for the population lower sensitivity bound $L(\theta)$, computed from the sample:

$$\ell_n(\theta) = \widehat{L}_n(\theta) - \frac{\widehat{q}_{L,1-\alpha}(\theta)}{\sqrt{n}},$$

where $\widehat{q}_{L,1-\alpha}(\theta)$ is the $1-\alpha$ quantile of the recentered subsampling distribution.

Let $\underline{\theta}_n = \inf\big(\{\theta \in [0,1] : \ell_n(\theta) \le 0\} \cup \{1\}\big)$, i.e., the first sensitivity value where the estimated lower confidence limit becomes nonpositive, or 1 if it never does.

Assume that $\ell_n(\theta)$ has a pointwise asymptotic coverage of at least $(1-\alpha)$. The first nonpositive value of ℓ_n, capped at one, is a lower confidence bound $\underline{\theta}_n$ satisfying

$$\liminf_{n\to\infty} P(\underline{\theta}_n \le \theta^\star) \ge 1-\alpha.$$

Proof. If $\theta^\star = 1$, then the probability statement is $1 \ge 1-\alpha$ for any $\underline{\theta}_n$.

Suppose, so, $\theta^\star < 1$. Then continuity and initial positivity give $L(\theta^\star) = 0$. On the coverage event, $\ell_n(\theta^\star) \le L(\theta^\star) = 0$, so $\underline{\theta}_n \le \theta^\star$. By the assumed pointwise coverage, this event has probability with limit inferior at least $1-\alpha$.

Generate a fixed grid $0 = t_0 < \cdots < t_K = 1$. Let $k \ge 0$ be the first index with $\ell_n(t_k) \le 0$. Return t_{k-1} if $k > 0$, 0 if $k = 0$, and 1 if no such index exists.

For the grid implementation, let $t_j < \theta^\star \le t_{j+1}$, as the sensitivity L is nonincreasing at θ, $L(t_{j+1}) \le 0$. By Proposition 7, provided its assumptions hold at t_{j+1}, the event $\ell_n(t_{j+1}) \le L(t_{j+1})$ has probability with limit inferior at least $1-\alpha$. On this event, $k \le j+1$, so the returned value is at most $t_j < \theta^\star$. If $k = 0$, the returned value is zero. Thus the grid procedure also has asymptotic coverage of at least $1-\alpha$.

□

E Minimal code for the IV perturbation bounds

The following Python script requires libraries SymPy and pycddlib 3 with its exact rational backend. Run it as `python minimal_iv_bounds.py`.

The script prints lower and upper bounds for the programs that generate closed-form solutions for the monotonicity and exclusion perturbation functions, as described in Appendix C.2.1. The variable names `pya_z` denote $P(Y = y, A = a \mid Z = z)$, with `y`, `a`, and `z` replaced by their binary values. The variable `theta` is the exact violation proportion, and `w` is $P(Z = 1)$. The output was simplified by removing terms that are redundant and algebraically grouping the remaining expressions.

```
"""Run with Python, SymPy and pycddlib>=3; no autobounds import is needed.

Print exact perturbation bounds with all observed cells, theta and w symbolic.
Only evaluate them on the printed compatibility domain with denominator > 0.
"""
from itertools import product, combinations
from fractions import Fraction
import sympy as s
import cdd.gmp as cdd

p = s.symbols('p00_0 p01_0 p10_0 p00_1 p01_1 p10_1', real=True)
theta, w = s.symbols('theta w', real=True)
x = s.Matrix([1, *p, theta])
delta = p[0] + p[2] - p[3] - p[5]


def hull(points):
    V = cdd.matrix_from_array(
        [[Fraction(str(v)) for v in row] for row in points],
        rep_type=cdd.RepType.GENERATOR)
    H = cdd.copy_inequalities(cdd.polyhedron_from_matrix(V))
    assert not H.lin_set  # These coordinates have full affine dimension.
    return [list(map(s.Rational, row)) for row in H.array]


def bounds(exclusion):
    points = []
    for a0, a1 in product((0, 1), repeat=2):
        if exclusion and a0 > a1:
            continue
        for ys in product((0, 1), repeat=4 if exclusion else 2):
            y = ys if exclusion else ys + ys
            obs = [int((a0, a1)[z] == a and y[2*z+a] == b)
                   for z in (0, 1) for b, a in ((0, 0), (0, 1), (1, 0))]
            violation = int(y[0] != y[2] or y[1] != y[3]) if exclusion else int(a0 > a1)
```

```
            N = [int(a0 < a1)*(y[2*z+1]-y[2*z])
                 for z in range(2 if exclusion else 1)]
            points.append([1, *obs, violation, *N])
    d = 2 if exclusion else 1
    H = hull(points)
    domain = [s.Ge((s.Matrix([r])*x)[0], 0)
              for r in hull([v[:-d] for v in points])]
    c = s.Matrix([1-w, w] if exclusion else [1])
    pieces = {1: set(), -1: set()}
    for rows in combinations(H, d):
        A = s.Matrix([r[-d:] for r in rows])
        if not A.det():
            continue
        b = s.Matrix([r[:-d] for r in rows])*x
        for sign in (1, -1):
            lam = A.T.inv()*(sign*c)
            condition = s.reduce_inequalities(
                [v >= 0 for v in lam] + ([w >= 0, w <= 1] if exclusion else []), w)
            if condition == s.false:
                continue
            value = s.expand(-sign*(lam.T*b)[0])
            pieces[sign].add(s.Piecewise((value, condition), (-sign*s.oo, True)))
    denominator = delta if exclusion else delta + theta
    lower = s.Max(*sorted(pieces[1], key=s.default_sort_key), evaluate=False)
    upper = s.Min(*sorted(pieces[-1], key=s.default_sort_key), evaluate=False)
    return lower/denominator, upper/denominator, domain, denominator


if __name__ == '__main__':
    for exclusion in (False, True):
        lower, upper, domain, denominator = bounds(exclusion)
        print('\nEXCLUSION' if exclusion else '\nMONOTONICITY')
        print('Require:', domain, 'and', denominator, '> 0; 0 <= w <= 1')
        print('Lower:', lower)
        print('Upper:', upper)
```